

STEERING THE FUTURE OF ADVANCED MATERIALS: STRATEGIC INTELLIGENCE IN ACTION

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Abstract

Policymakers aiming to foster and govern emerging technologies may wish to develop anticipatory capacities and strategic intelligence to inform and future-proof their efforts. This report delves into how these approaches are being mobilised, using advanced materials as a policy-relevant case given their regular inclusion in critical emerging technology lists and the rise of national strategies to promote their research to meet strategic goals. A series of case studies from across the globe are brought together to illustrate practical applications in this field of anticipatory coordination efforts, upstream governance and “by-design” approaches, and strategic intelligence tools. Insights from this report aim to support policymakers and technology assessment practitioners seeking to develop or update national strategies for emerging technologies like advanced materials to accelerate innovation and further develop national ecosystems.

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- National advanced material strategies,
- Anticipatory processes that inform national strategies, such as roadmapping and stakeholder engagement activities, upstream governance and “by design” approaches,
- The strategic intelligence tools and approaches that are mobilised in the anticipatory processes mentioned above.

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Executive Summary

Advanced materials (also referred to as “novel materials”, “value-added materials” or “materials innovation”) are **regularly listed as a critical emerging technology**, encompassing highly diverse applications across multiple sectors – from healthcare to defence to chemical manufacturing. Whilst this varied nature means there is no global standard definition, the OECD Working Party for Manufactured Nanomaterials (WPMN) has developed a working description of materials designed with new or enhanced properties, or targeted structural features to achieve specific or improved performance. These attributes make them **essential to foster innovation in many sectors, drive economic growth and help tackle societies’ grand global challenges**.

Recognising the current role and future potential of advanced materials to pursue strategic goals, **governments are increasingly investing in policy tools** to support their development and implementation, **most notably via dedicated national strategies**. In constructing these, governments gain a unique opportunity to map out the sector’s state of play (e.g. national capacities), understand key associated factors (e.g. socioeconomic impacts, global landscape), and identify their country’s unique challenges and opportunities. Strategies can then set a desirable vision for the future of the national advanced materials ecosystem around which actors can coalesce in pursuit of strategic goals.

While different policy drivers will shape these efforts depending on the national context, this report identifies a number of common themes. For example, advanced materials are regularly identified as essential elements to **ensure resilience, particularly with a view to issues around access to critical resources** to support economic and national security – in the face of unreliable supply chains and new trade barriers. Advanced materials are a key area driving innovation in the defence sector. Advanced materials are a powerful **engine for economic growth** and employment across sectors: many countries set out strategies with the explicit aim of securing or maintaining **global leadership** in the space to provide a competitive advantage in global markets, foster technology security and allow them to shape the technology in their desired direction. They will also be foundational in the digital transformation of advanced manufacturing industries. They are also considered to be key **enabler for the green transition**, and concurrently to adapt to new international trade rules setting sustainability standards.

The development of these national strategies mobilises a number of anticipatory processes, some of which are illustrated in this report through a series of case studies from across the globe, including some deeper dives. This analysis is divided into three sections.

The first focuses on **anticipatory coordination efforts**, which aim to use strategic intelligence producing processes as a way to bring together the various stakeholders from across the value chain to coordinate a path to a functioning new innovation ecosystem. It includes activities such as roadmapping, participatory foresight and collective agenda building.

The second focuses on **upstream governance and “by-design” approaches**, which aim for proactive identification of information gaps, concerns, and regulatory needs early on, and thus lay the groundwork for research, guidelines, testing systems, and legislative action to mitigate or avert potential negative

impacts. It includes principles like anticipatory agile and adaptive governance, life cycle assessments, or the OECD's Safe(r) and Sustainable Innovation Approach (SSIA).

The third relates the use of '**strategic intelligence**': the analysis and knowledge of possible directions and economic stakes of technology development, the possible ethical and societal aspects, and the potential impacts. This report interrogates the use of various strategic intelligence tools and approaches – from horizon scanning and technology forecasting to text mining – across the case studies to better understand their role, how and when they are used, and in which combinations.

Through these insights, **this report aims to inform policymakers seeking to develop or update national strategies for new advanced materials** to accelerate innovation, develop national ecosystems and implement solutions. By analysing the anticipatory tools employed in existing activities and how strategic intelligence is generated and used, it also aims to **support technology assessment practitioners in providing timely intelligence** to their audiences to steer the direction of technology development. Future work could assess the quality and outcomes of both the anticipatory processes and the strategic intelligence activities, as well as examine the local ecosystems of organisations producing strategic intelligence that the national strategies can draw upon.

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1 Introduction

The unique properties and rapid innovation of advanced materials hold great promise to build safer, more sustainable and higher performing products for more resilient societies and to tackle some of the critical crises of our times. For example, new photovoltaic materials and batteries could help further harness the potential of renewable energy, and bio-based products could diminish the need for fossil fuel feedstocks and usher in a circular economy. Beyond this, advanced materials could usher in innovations in sectors as diverse as aerospace, food packaging and soft robotics (ChemistryNL, 2021^[1]).

Whilst advanced materials are increasingly seen as a critical area of emerging technologies, there is no global standard definition of advanced materials, and national strategies on the subject often use broader terms such as “materials innovation”, “novel materials” or “value-added materials”. Agreeing with the OECD Working Party for Manufactured Nanomaterials (WPMN) the term “advanced materials” is often used broadly and in different contexts, the Steering Group mobilised the “working definition”¹ of Advanced Materials by the WPMN, to provide clarification in terms of scope of this project activity:

Advanced materials are understood as materials that are rationally designed to have (i) new or enhanced properties, and/or (ii) targeted or enhanced structural features with the objective to achieve specific or improved functional performance. This includes both new emerging manufactured materials (high tech materials), and materials that are manufactured from traditional materials (low tech materials). This also includes materials from innovative manufacturing processes that enable the creation of targeted structures from starting materials, such as bottom-up approaches. It is acknowledged that what are currently considered as advanced materials will change with time.

The report has been built on three “data rich” sections:

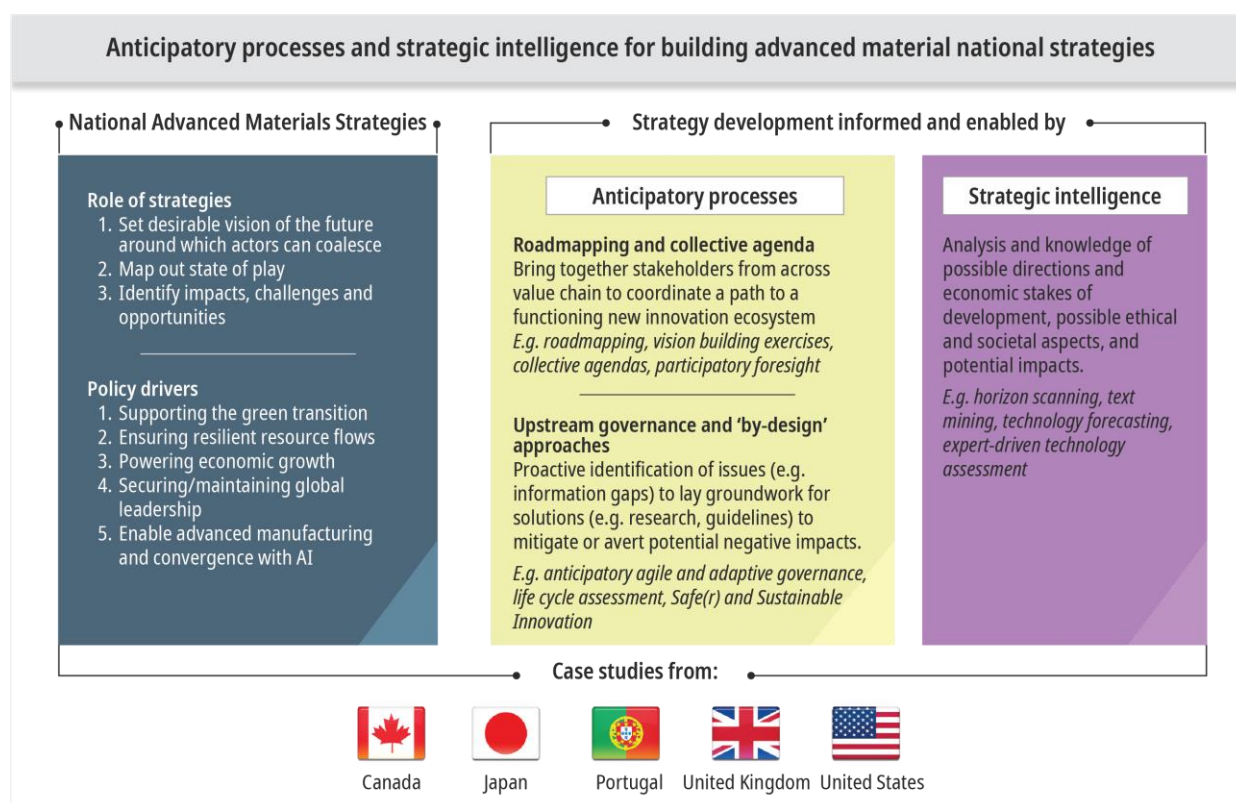
1. **National advanced material strategies** – This section analyses **advanced material strategy documents** (or advanced material related areas like advanced manufacturing) to articulate the underlying policy drivers embedded in these strategies.
2. **Anticipatory processes that inform, or enable, national strategies** – This section looks at activities and processes that facilitate the building of new advanced material value chains and innovation ecosystems (through examples of roadmapping activities, coordination programmes and anticipatory actions in research and innovation cluster activities) as well as lead to good governance of early-stage emerging advanced materials (through upstream governance and “by design” approaches like anticipatory Life Cycle Assessments and Safe and Sustainable by Design).
3. **Strategic intelligence roles and capacities** – This section makes explicit the strategic intelligence tools and approaches mobilised in the anticipatory processes mentioned above to identify the role and application of strategic intelligence approaches, and articulate the capacities needed for contemporary advanced material strategy development.

These are supplemented by three Annexes comprising the full list of advanced materials-related policies identified, the guiding questions to support experts in case study development, and the case studies provided by Steering Group participants.

Therefore the outcomes of this project activity, stemming from the three pillars, provide key insights into forward-looking advanced material strategies and cases of collective strategic intelligence production *in action*. More specifically, the outcomes include: (a) an insight into the policy priorities behind national

advanced material strategies and how they are framing future policy levers in the field, (b) an understanding of anticipatory processes such as roadmapping and industrial vision building across different countries, as well as the co-development of anticipatory governance mechanisms of advanced materials, particularly through “by design” approaches, and (c) an articulation of the roles and applications of strategic intelligence approaches, and the capacities needed for contemporary advanced material strategy development.

Figure 1.1. Summary of key report insights



Source: OECD Research

2 National advanced material strategies

Introduction

Demand for advanced materials is growing and is expected to continue doing so in the future, in particular for technological solutions in renewable energy, semiconductors, medicines and defense applications (European Commission, 2024^[2]). However, traditionally, advanced material innovation has a relatively long development timeframe – 10 to 20 years from inception to market launch – meaning that solutions may be much delayed and may be unable to meet current urgent demand. This situation requires policy investments and renewed efforts to accelerate the translation from research laboratories to market, including the active support of emerging supply and value chains, as well as the accompanying framing conditions (technical standards and other forms of governance) (AMI2030, 2023^[3]). This issue is increasing in importance for governments and is reflected in advanced materials being included in several national critical technology lists and dedicated national strategies. Given differing national contexts and priorities, each country has its own constellation of drivers for this push, but there are common themes in both the goals they pursue and how they aim to achieve them.

Governments may **establish a desirable vision of the future, around which actors can coalesce** in pursuit of strategic societal goals (e.g. maintaining competitive global leadership, economic growth, drive to net zero carbon emissions), addressing crises (e.g. loss of biodiversity), strengthening sectors (e.g. defense and national security) or guiding development and deployment of critical technologies (e.g. artificial intelligence, biotechnology).

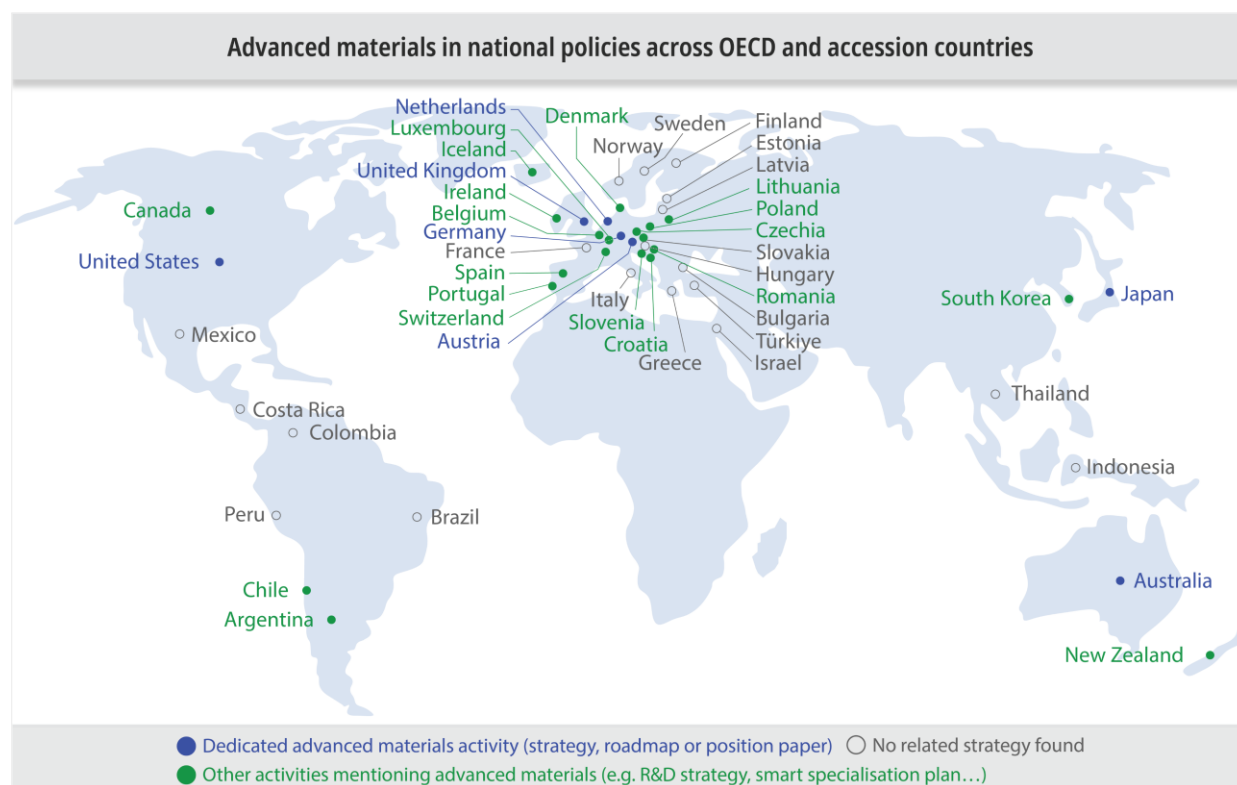
The process of constructing a national vision and associated strategy allows nations to **map out the state of play** for sectors, technologies or challenges. Understanding key factors like current national capacities, socioeconomic impacts, and the global competitive landscape, can tease out the areas of highest potential impact for a country as well as its key assets. At the same time, this mapping can be used to **identify the challenges in achieving the desired vision**, which could include dependence on other countries for resources, gaps in national infrastructure, barriers to scale-up and commercialization – to name a few.

Landscaping Advanced Material Strategies in OECD countries

To understand differing national approaches to fostering the domestic advanced materials sector, the OECD Secretariat has mapped related national activities in OECD member states and accession countries, primarily via desk research and complemented by input from a dedicated expert group. In some countries – particularly those with vibrant advanced materials ecosystems like Japan and the United States – **governments have dedicated national strategies or roadmaps for promoting advanced materials**, identifying ambitious goals and initiatives to achieve them. Other countries still recognize the strategic and scientific importance of the sector, but instead of developing specialized plans they have **included**

advanced materials into broader strategies, such as R&D roadmaps or smart specialization plans. These insights are detailed in Figure 2.1. and Table 1 in the Annex.

Figure 2.1. National strategies, roadmaps or positions mentioning advanced materials across OECD member states and accession countries

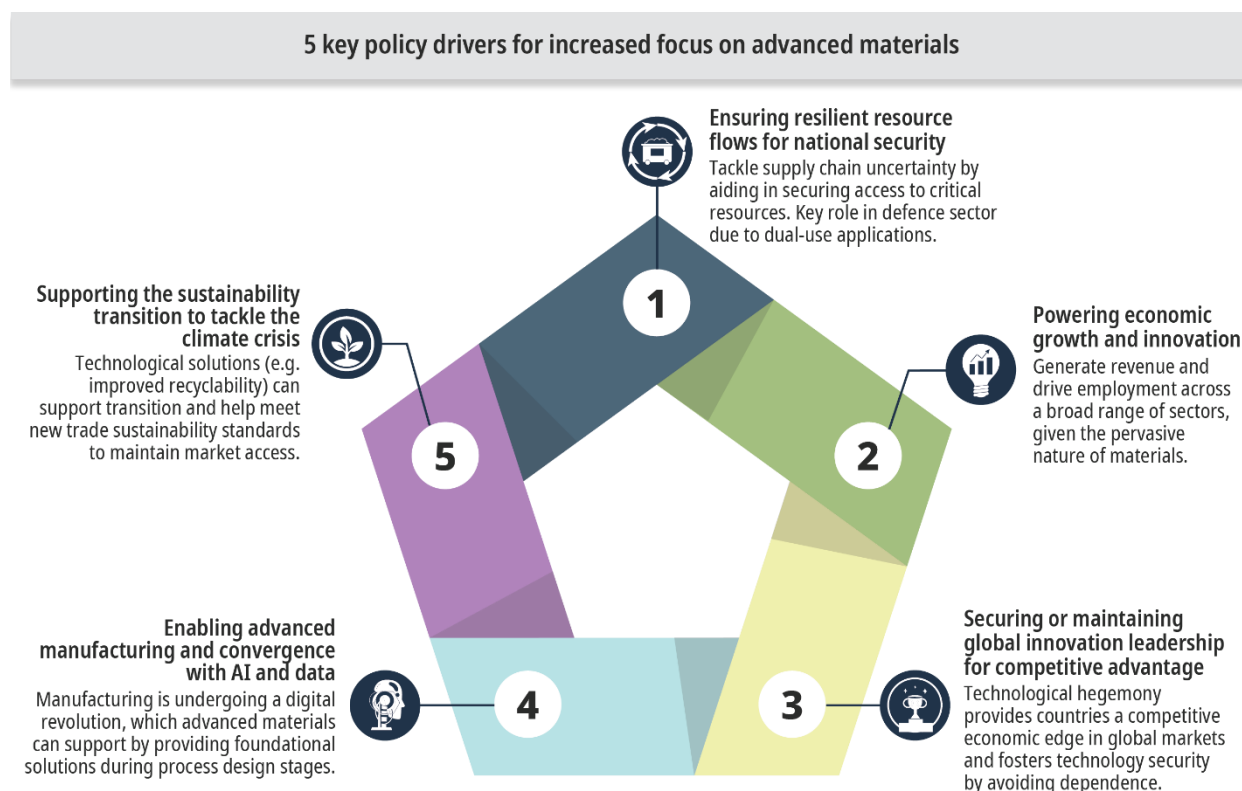


Note: Only OECD countries and candidate accession countries (as of November 2024: Argentina, Brazil, Bulgaria, Croatia, Indonesia, Peru, Romania, Thailand) were investigated. All other countries are left blank.

Source: OECD Research.

Policy drivers for increased focus on advanced materials

Although the policies identified above respond to unique national contexts, they can be compared and contrasted to identify common themes and global trends on why countries are developing strategies, what they are aiming to achieve and what approach they are following. These are listed below in no particular order, as different countries will prioritize these drivers differently.

Figure 2.2. 5 key policy drivers for increased focus on advanced materials

Note: These are shown in no particular order, as different countries will prioritize these drivers differently.
Source: OECD Research.

Policy Driver 1 – Ensuring resilient resource flows for national security. As geopolitical uncertainty increases, unreliable international supply chains combined with new trade barriers are challenging access to critical raw materials concentrated in limited sources. This access is not just important from an economic competitiveness perspective, but can also be key for national security, as some assets are becoming increasingly essential to power and protect national socio-economic activities – for example, vital health resources like medicines or technological components like semiconductors. There is an increasing need to secure vital resources and explore which advanced materials could support strategic autonomy. For example, advanced materials can improve durability and recyclability, help develop processes that utilize more readily available alternatives, or even outright replace some of these critical raw resources (AMI2030, 2023^[3]).

A sub-element of this policy driver is advanced materials' key role for innovation in the **defence sector**, due to dual-use applications. Advanced materials can be stronger, more durable and lighter than conventional materials (which can improve resistance and fuel efficiency), whilst new properties like temperature resistance, self-repair or stealth capabilities can provide revolutionary applications (Burnett et al., 2018^[4]). Advanced materials are also essential for increasingly critical sectors like space technologies and satellite communications (MaterialsNL, 2021^[5]).

Policy Driver 2 – Powering economic growth and valuable innovations. Scientific and technological innovation serve as a key economic motor for countries, creating jobs in research and manufacturing and generating revenue. For example, in the United Kingdom in 2023, the materials-related manufacture and export generated £430 billion in revenue, employed over 2.6 million people and accounted for 49% of all UK exports (Henry Royce Institute, 2024^[6]). Such innovations also represent a long-term investment into

society, as materials discoveries made last century are still providing revenue today. Furthermore, the pervasive nature of materials means they can impact practically all sectors: beyond the aforementioned sustainability and defence fields, materials innovation can support advancements in healthcare, agriculture, construction and electronics, amongst others. For instance, smart materials could act as biosensors for diagnostics or 3D print organs and body parts, membranes to purify water could increase access to clean sources (MaterialsNL, 2021^[5]).

Policy Driver 3 – Securing or maintaining competitive advantage on global markets. As more countries recognize the importance of advanced materials for sustainability, economic competitiveness and national security, and invest accordingly, the global landscape of innovation is shifting. Emerging countries in particular are driving this change, supported by factors like cheaper labour and resource costs and the advent of digitalisation. Historically-leading nations in advanced materials research – such as Japan, Germany, the United Kingdom and the United States – are facing growing competition, exacerbated by the commodification of products (involving faster spread, shorter lifecycles and more competitive pricing) whilst concurrently requiring more complex products at a time of challenging access to resources due to unstable supply chains (Council for Integrated Innovation Strategy, Cabinet Office, Government of Japan, 2021^[7]). The desire to maintain or obtaining **technological hegemony** is a strong driver, as not only does it provide countries a competitive economic edge in global markets, but it also fosters technology security by avoiding dependence on others. In addition, being at the tech frontier allows them to shape technology development according to their values and wield influence in international discussions to shape the future of the field.

Policy Diver 4 – Enabling advanced manufacturing and convergence with AI and data. Manufacturing is undergoing a revolution, underpinned by new opportunities provided by data analytics and artificial intelligence. This is in response to increasingly complex supply chains calling for multidisciplinary integration and customization of products and processes, as well as to lower labour costs and meet high demand rates. These developments are reshaping the global manufacturing landscape and enabling novel countries to participate in new ways. Scientific and technological advancements are identified as key to competitiveness in this changing space, and although many enablers will help foster this transition (including but not limited to sensors, robotics, 3D printing and augmented reality), developments in advanced materials are identified as a key pillar. At the moment, their unique properties are used to help tackle product-specific issues, such as weight or durability, but in the future they could be incorporated as early as the design stage to speed up the development process and provide more foundational solutions (CSIRO, 2016^[8]).

Policy Driver 5 – Supporting the sustainability transition to tackle the climate crisis. Developing more circular and sustainable societies – to tackle the climate crisis, increased pollution and loss of biodiversity – is an urgent task forcing practically all sectors to evolve. Advanced materials hold promise in supporting this transition by providing novel technological solutions. For example, they can increase a product's durability or their recyclability for a more circular economy; be used to produce renewable energy to replace fossil fuels and reduce carbon dioxide emissions; and catalyse processes for increased efficiency or reduction of waste. Recognising this potential, the German Federal Ministry of Research, Technology and Space has established “Mat2Twin” - the largest German funding program for material sciences at national level - to foster research on materials for the green and digital transformations, including the creation of more closed-loop systems and recycling of high-value resources (Federal Ministry for Economic Cooperation and Development, 2025^[9]). However, advanced materials can also lead to environmental harm and thus their own safety and sustainability should be ensured throughout their life cycle to maximise the opportunities provided by this technology. The German Environment Agency, for example, advocates for following the precautionary principle and identifying concerns for sustainability (e.g. toxicity, increased ecological footprint) at an early stage to close knowledge gaps early and re-design when necessary (Schwirn and Völker, 2023^[10]).

A sub-driver relates to **new trade sustainability rules**. As countries around the world implement policies to drive sustainability, it may be the case that other nations wanting to remain competitive in the international trade space will have to ensure their exports meet sustainability standards to maintain access to current markets (Council for Integrated Innovation Strategy, Cabinet Office, Government of Japan, 2021^[7]). A prime example is the European Union's Carbon Border Adjustment Mechanism, which aims to ensure the price of carbon in imports is accounted for and equalled to that of domestic products to prevent offshoring of emissions². Countries wishing to maintain competitive access to the EU market would need their products to be sustainable and produce low emissions, which advanced materials could enable.

Adequately responding to these drivers to achieve the underlying aim – fostering a strong advanced materials innovation and manufacturing space – **requires action from a broad range of stakeholders**, from academics pursuing fundamental research to industry deploying practical solutions. Each of these players have different capabilities to act and do so in response to unique incentives. For example, academics may decide to pursue further research due to new funding opportunities, whilst industry players may invest more in expectation of greater return on investments. **Governments, and associated public agencies, can play a role in translating these trends and drivers into concrete policies and incentives** to foster concerted action by different stakeholders.

3 Anticipatory processes that inform, or enable, national strategies

Roadmapping and collective agenda building activities

Introduction

Anticipatory coordination is a term that encompasses all activities and fora designed to bring together the various stakeholders needed to power and drive advanced material value chains and innovation ecosystems. Acting together, either through direct collaboration or in full knowledge of each other's strategies, ecosystem actors coordinated to achieve a path to a functioning new innovation ecosystem or value chain.

The most well-known anticipatory coordination mechanism is the International Technology Roadmap for Semiconductors³, that set the pace for development and performance improvement in the diverse ecosystem of specialized component and process manufactures – this coordinates the diverse members of the ecosystem and sets the collective pacing of these activities. By creating a regular roadmap, the pace of change is agreed to and suppliers can better align their development strategies with the overall vision and milestones outlined in the roadmap. The roadmapping process, therefore, not only **collectively envisions the future** status of the semiconductors innovation ecosystem (i.e. all of the different component developers and their supply chains) but **also sets the pace for the innovation chain**, collectively agreeing when certain new developments should be in place.

For advanced materials, the innovation ecosystem is much more fragmented and distributed – given that advanced materials are both diverse in their nature and are themselves enabling technologies, connecting to a variety of value chains and sectors.

Strategic intelligence production processes for anticipatory coordination can come in a variety of forms, such as vision building exercises, participatory foresight and roadmapping activities, to name a few.

Selected country examples on anticipatory coordination processes

United Kingdom's National Materials Innovation Strategy

Facilitated by the Royce Institute, the Strategy outlines the vision of the UK's **world-leading advanced materials ecosystem driving economic growth and tackling grand challenges**. Since the materials sector is crucial for the UK's manufacturing output, (contributing significantly to employment, exports, and R&D investments), the Strategy calls for substantially enhancing the value of UK-based industries in the rapidly expanding multi-billion-pound materials markets. Recognizing as well that advanced materials are integral to addressing societal and technological challenges like achieving sustainability goals, the strategy also emphasizes the necessity of materials innovation in achieving a prosperous, net-zero economy.

The Strategy report⁴ highlights a **demand-led approach (application-centric, rather than material-centric)**, emphasizing collaboration between government, industry, and academia to identify and **prioritize materials innovations that align with national and industrial needs**. Insights were drawn from forward-looking Expert Working Groups that conducted deep dives into core themes – such as energy materials, biocompatible materials, and sustainable soft materials – to **identify key innovations and enablers required for commercialization and scale-up**. This collective strategic intelligence fed into the development of the national strategy for the UK advanced materials innovation system. More details can be found in Annex B.

Portugal's Industry & Manufacturing Agenda 2030

This forward-looking exercise set out to evaluate the potential short- and long-term impacts of emerging technologies on the Portuguese industry, including advanced materials as one of the five priority domains selected by the experts engaged. It was orchestrated to create strategic intelligence to (a) identify the key target areas for the Portuguese manufacturing ecosystem and (b) **mobilise and build cohesion across the ecosystem in advance**. Experts did so by determining the key challenges to national innovation (e.g. incorporating new knowledge and technological skills into firms, integrating Industry 4.0 and circular economy concepts) and the global trends shaping the national industry (e.g. climate change, resource competition, rising social responsibility, and digitalization's impact on manufacturing and business models).

By **establishing shared goals and timelines**, this collaborative agenda-setting process provided strategic guidance for developing science, technology, and innovation policies in Portugal, and inform coordinated development efforts for government, industry and academic stakeholders. More details can be found in Annex B.

Canada's PRIMA Quebec

Canada's Quebec province created PRIMA Quebec in 2014 in recognition of the strategic importance for its manufacturing sector of supporting the local advanced materials ecosystem. It has developed studies that have provided key strategic intelligence: categorizing and mapping of the province's industrial ecosystem; identification of international trends likely to stimulate demand; end markets to reveal challenges and opportunities throughout value chains; and local demand by surveying industry associations and research consortia. In September 2024, PRIMA Quebec completed an 'Advanced Materials 2035 Roadmap' exercise that convened the advanced materials ecosystem to **mobilize and increase the circularity of advanced materials within supply chains to advance the decarbonization** of the Quebec economy.

PRIMA Quebec's work has helped drive policy, **providing evidence for both provincial and federal innovation agencies to create funding programs** to address challenge and gap areas identified in their reports. Its role as convener of industry, academia and government has been critical to driving collaboration amongst actors, and has also been informative for industry players by providing connections, funding, talent and supporting research organizations. More details can be found in Annex B.

United States' Advanced Materials and Manufacturing Technologies (AMMT) Roadmap (2022)

An initiative by the US Department of Energy's Office of Nuclear Energy (DOE NE), it has the targeted aim to support the **speedier transition of new advanced material technologies into nuclear systems** to maintain US leadership in the field. This overarching vision is broken down into more concrete goals, related to **accelerating the technology's development, qualification, demonstration, and deployment** – such as establishing a framework for rapid qualification or bringing new materials to market

in a timely manner. The roadmap identifies key research needs, opportunities and challenges in the short- and long-term, outlining strategic research priorities and providing a five-year plan to realize its aims.

Each of the goals has a program element which leverages and integrates existing programs within the DOE NE, but also **promotes cross-government collaborations** with other agencies like the Office of Science, Department of Defence, NASA, etc. as well as independent research institutes like the Electric Power Research Institute and Nuclear Energy Institute which work on advanced materials and have industrial partners. The AMMT leadership team and its steering committee represent different national labs, whilst an industry council will be established in the future.

United States: U.S. Government Accountability Office Technology Assessment of Critical Minerals Recovery Technologies

Another example of providing strategic intelligence for anticipatory governance is a technology assessment on critical minerals that was prepared by the U.S. Government Accountability Office (U.S. GAO) to assist Congress with its oversight responsibilities.⁵ This report built upon a history of government actions, including a December 2017 President Executive Order which outlined the country's strategy to ensure supplies of critical minerals and the creation of a national strategy.⁶ A **national strategy which outlined actions that federal agencies should take to diversify and expand critical mineral supplies** obtained from sources other than traditional mining was created in June 2019.⁷

A performance audit conducted from February 2021 to June 2022 by the U.S. GAO found that the national strategy did not fully incorporate certain characteristics of effective national strategies, including what the strategy intends to achieve, how results will be achieved, or performance measures to gauge results; what the strategy will cost; and how federal agencies will implement it. Accordingly, the U.S. GAO recommended that the three cognizant federal agencies update the 2019 critical minerals national strategy as it relates to recovery and substitution and include (1) **goals, subordinate objectives, activities, and performance measures**; (2) **resources, investments, and risk management**; and (3) **integration and implementation**. As of May 2025, these recommendations remained open.⁸

In 2024, the U.S. GAO issued a technology assessment that examined technologies for recovering critical minerals for key domestic, nontraditional sources that could be used to recover critical minerals and the associated benefits and challenges. It also identified three policy options – possible actions by Congress; federal agencies; and/or state, local, and tribal governments – that could help address the challenges or enhance the benefits of recovering critical minerals from nontraditional sources.⁹ More details can be found in Annex B.

Japan's Materials Commercialization Innovation and Cultivation Ecosystem Strategic Innovation Promotion Program (SIP)

The national cross-ministerial Strategic Innovation Promotion Programs (SIP) aims to address Japan's major social problems and revitalize the economy by focusing on end-to-end R&D. The 3rd SIP period was launched in 2023 with 14 projects aiming to realise "Society 5.0". The framework-building process (organized by the Cabinet Office's Secretariat of Science, Technology and Innovation Policy and the SIP governing board) involved **back-casting from the vision of "Society 5.0" to identify target areas** for promoting end-to-end R&D, incorporating opinions and issues arisen during the mid-term evaluation of the 2nd SIP. Given advanced materials' importance to Japan as a fundamental technology, one of the projects was "*Creating a materials innovation ecosystem for industrialization*" which plans to develop a data-driven development infrastructure to develop materials companies and promote societal implementation.

Goal scenarios for **five readiness levels indicators (technology, governance, business, social acceptability, and human resources)** were defined, and by back-casting from these, several readiness levels were set for each indicator. A roadmap was formulated based on the desired target levels to be

accomplished during the 3rd SIP period, in three phases: building the base of the ecosystem (2023-2027), developing the ecosystems via practices (2028-2032) and establishing circular growth (2032-2040). Macro-trend analysis for targeting potential markets were also considered.

Feasibility studies were carried out by program directors in collaboration with related ministries and the project management institutes. The results informed the establishment of an R&D plan which was assessed through public comment procedure. Throughout the process, perspectives from stakeholders such as economic organizations, industry, academia, think tanks and citizens were included. More details can be found in Annex B.

European Union's Communication on Advanced Materials for Industrial Leadership (2024)

The Communication **aims to maintain the bloc's industrial leadership and leverage the technology's potential to drive resilience and the green transition**. It has three overarching aims: foster R&D, scale-up innovation and manufacturing capacity, and accelerate industrial uptake and market deployment. By identify the challenges the EU faces in the space – such as fragmented research ecosystems across member countries, insufficient private funding, or lack of skills and standards – it sets out further steps in 5 pillars, each with clear and measurable actions for the Commission, member states and industry.

First, the development of a joint strategic European approach across member states, which determines common research priorities to foster cross-border collaboration. Second, improving scale-up and manufacturing capacity via new data sharing infrastructures to foster digitalisation or improving access to testing facilities. Third, increased financing by leveraging existing funding tools or developing new ones, like public-private partnerships. Fourth, supporting production and use of advanced materials, with actions such as developing common industry standards, addressing patenting challenges, increasing skilling programs and considering public procurement.

Fifth is establishing a governance framework, recognising the need for key players to steer the implementation of the Communication's proposed actions. To do so, a **Technology Council for Advanced Materials will be set up with representatives from Member States, academia, industry, and other key actors**. It will support the efforts listed above, as well as align efforts at different levels of government, explore international partnerships and consider monitoring results.

Previous initiatives could have built momentum for this more institutional action. The Advanced Materials Initiative 2030, made of representatives from research institutes and industry, developed a Strategic Materials Agenda in 2023, whilst the European Commission Expert Group on advanced materials, which aims to deepen exchange of information between member states and the Commission within Horizon Europe, has been active since 2014.

Upstream governance and “by design” approaches

Introduction

Developing novel advanced materials brings uncertainties about their safety, sustainability, and the adequacy of existing regulations. Early detection of these issues is essential for regulators and innovators to prepare for potential actions. Such a **proactive approach helps identify information gaps, concerns, and regulatory needs early on**, laying the groundwork for relevant research, guideline development, appropriate testing systems, and legislative updates to mitigate or avert potential negative impacts. This could help prevent uncertainties in risk assessments from resulting in unsafe products being released into the market, potentially leading to public backlash, reduced industrial investment hampering innovation, or the need to urgently replace harmful substances.

Effective innovation policies, in addition to catalyzing and driving novel advanced material development, can ensure **robust governance approaches that keep the regulation of potential risks at pace with innovation**. The timely identification of potential health, environmental, and sustainability impacts is not only necessary to comply with policies and strategic visions to foster the green transition but can also enhance the innovation process by making it more responsive to the reality and needs outside of the lab.

Anticipatory capacities and agile and adaptive governance are key, as outlined in the recent [OECD Framework for Anticipatory Governance of Emerging Technologies](#), as forward-looking approaches can provide a means of assessing existing risk governance approaches in a situation of paucity of data. Emerging risks include new risks (stemming from novel materials or technologies) or familiar risks that are becoming apparent in new or unfamiliar conditions. Such risks are highly uncertain and often have limited evidence on which to base assessments. At the same time, it involves **not only the anticipation of potential risks and benefits, but also the exploration how these are perceived and framed by different stakeholders**. Such framing is dependent on the values of those perceiving and thus unpacking what are the underlying norms and values around an emerging technology and its possible impact is key. One example is the International Risk Governance Council (IRGC) guidelines¹⁰ for emerging risk governance: a set of flexible guidelines that support public and private organisations in designing internal processes to proactively deal with emerging and evolving risks.¹¹ Another challenge observed is that current regulations may not adequately address the specific challenges posed by advanced materials, requiring updated and agile regulatory frameworks founded on anticipatory governance.

Selected examples for anticipatory governance of advanced materials

European Commission: Safe and Sustainable by Design framework

Life Cycle Assessment (LCA) principles are increasingly being adopted by innovators aiming to improve their sustainability, as these encourage consideration of environmental and social impacts throughout a product's life cycle, beyond the design and production stages. As uptake of these tools increases and they become more sophisticated, specialised methods are arising, such as Environmental Life Cycle Assessments (E-LCA) that measure the extraction, consumption and release of resources like energy, water and gases, as well as Social Life Cycle Assessments (S-LCA) that measure positive and negative social impacts on affected stakeholders. These quantitative approaches to measuring impact provide innovators additional insights that can inform design choices for new products or identify “hotspots” in a life cycle which can be addressed for better environmental impact (KPMG, 2023^[11]).

LCA can be complemented by following a **Safe and Sustainable by Design (SSbD)** approach, which aims to ensure that new materials are not only high-performing but also sustainable and safe throughout their life cycle. This involves integrating safety, environmental impact, and circularity considerations – such as those gleaned from LCAs – from the early stages of material design and development into production and end-of-life. Specific focus areas include designing circularity in polymers and elastomers, developing biodegradable and recyclable materials, and creating materials that reduce reliance on toxic or scarce raw materials.

A practical example is the European Commission's 2022 safe and sustainable by design (SSbD) framework, which provides companies and member state authorities practical guidance on the design and assessment of chemicals and materials. It evaluates the material's hazardous properties and the impact on human health and the environment in the production, use and end-of-life phases. These insights can be used to develop novel products or redesign existing ones with improved safety and sustainability at all or key stages. The framework is currently being tested by innovators, but is aiming to foster innovation, alternatives to substances of concern and become a global reference (European Commission: Directorate-General for Research and Innovation, 2022^[12]).

OECD: Early4AdMa early warning system

Equally necessary for innovations to reach the market is regulators' ability to effectively evaluate their novel properties. This can be challenging as regulatory knowledge and risk governance tools, given their need to be extensively validated, may struggle to keep pace with innovation. The concept of **Regulatory Preparedness (RP)** calls to improve regulator's capacity to anticipate technological developments and have the knowledge and tools for effective risk assessment. This could involve information sharing between innovators and regulators from early in the design process, under trusted environments (confidential conditions that protect commercial interests). (Soeteman-Hernández et al., 2019^[13])

The concepts of Safe and Sustainable by Design and Regulatory Preparedness have been brought together to form the **Safe(r) and Sustainable Innovation Approach (SSIA)**, whose aim is to reduce the time between a technological innovation emerging and the development of suitable risk assessment tools and frameworks. This concept has been operationalised in the **OECD's Early4AdMa¹² system**. Existing screening methods for novel materials primarily focus on safety and often neglect sustainability but the unique properties of advanced materials necessitate distinct safety and sustainability assessments compared to general chemicals. Recognising that there is a need for a screening approach that addresses this in a fast-evolving technological field, the OECD Working Party for Manufactured Nanomaterials developed an anticipatory screening approach labelled the Early4AdMa system as an agile and adaptive approach to identify both safety and sustainability concerns and to inform governance approaches.¹³

Box 3.1. Example: OECD's Early4AdMa

Advanced materials offer many technological solutions but may also pose unforeseen risks towards the environment or human health. Anticipating these issues can help ensure safety and sustainability early in the innovation process and foster innovation. Towards this goal, a new **early warning system (Early4AdMa)** has been developed by the OECD Working Party for Manufactured Nanomaterials (WPMN) **to systematically identify and prioritise such risks**. By also providing tailored follow-on actions to decisionmakers and innovators, it gives them the opportunity to **anticipate and mitigate issues in a timely manner**, promoting Safe and Sustainable by Design (SSbD) principles.

Early4AdMa's methodology is divided into two tiers, each with several steps. Tier 1 starts with 1) scanning the field of innovative advanced materials, selecting a set of materials for further research, 2) screening the group by **NESSI (Novelty, Exposure, Severity, Scope, Immediacy)**, **as well as for sustainability concerns and regulatory applicability** and based on the resulting score, 3) a preliminary warning is developed explaining any potential concerns.

Tier 2 continues with 4) collecting additional information, based on the preliminary warning and to feed into 5) **scoring by a multidisciplinary and multistakeholder group of experts** on four major topics (safety assessments for human health and for the environment, regulatory applicability and sustainability). Each topic is broken down into sub-questions and the uncertainty is included in considerations. This is followed by 6) an assessment and prioritisation of the scoring results, **determining follow-up actions** (e.g. generate additional safety data to reduce uncertainty, share insights with regulatory agencies, ensure recyclability...) which are then **communicated to decision makers**. After enough time, experts would be asked to 7) reflect and evaluate the impact of the warning and the actions proposed.

Originally proposed by the Netherlands and Germany¹⁴, Early4AdMa was brought to the OECD Working Party for Manufactured Nanomaterials (WPMN) Steering Group that published the approach in September 2023, with an accompanying Excel table to support the scoring exercise. A previous iteration

of the approach was used for two case studies during OECD WPMN workshops and the feedback was incorporated into the published approach.

Source: (OECD, 2023^[14]).

Effective regulatory governance of advanced materials should recognize their unique properties. Existing risk assessment test methods developed for traditional chemicals may at times not be appropriate and adapted or novel ones customized to advanced materials needed. This requires extensive regulatory research and efforts to internationally harmonize ensuing protocols to foster cross-border cooperation and provide certainty to innovators. Fostering FAIR data principles would increase the value of data available to support regulatory research. Incorporating diverse perspectives, from regulators to industry to academia, is essential to develop robust and practicable governance processes. A governance body tasked with facilitating such multistakeholder dialogues and coordinating diverse initiatives to avoid duplication of research would be of value (Groenewold et al., 2024^[15]).

Japan: Nanotechnology Business Creation Initiative (NBCI)

Carbon nanotubes (CNT) are a promising type of advanced materials with a broad range of applications, from batteries to transmission cables to health. However, regulating their use is difficult as their diverse structures and properties are hard to characterize and risk assess. Nonetheless, efforts have been underway in Europe to do so, with the European Commission revising their Classification, Labelling and Packaging regulation and calling for certain CNTs to be labelled as presumed carcinogenic.

The Nanotechnology Business Creation Initiative (NBCI) – a Japanese industry association **composed of more than 170 companies, and which collaborates with Ministries and national research institutes** – recognised the importance of CNTs for the Japanese industry and created a **Social Acceptance and Standardization Committee to disseminate information** about CNT evaluation in Japan and abroad. For example, it submitted comments to ECHA’s public consultation calling for more scientific evidence on CNTs’ carcinogenicity, harmonizing classification with international assessments, and developing risk management methods.

NBCI’s advocacy has included partnerships with other organisations: it cooperated with the Japan Business Council in Europe (JBCE) to propose alternative classification criteria and present additional scientific evidence, as well as signed a memorandum of understanding with the Nanotechnology Industries Association (NIA) to cooperate in responding to the European regulation.

NBCI will continue these activities to **shape legislation to realize both innovation and safety based on scientific evidence**. Specifically, it plans to execute further research on the possible mechanisms of CNT harmfulness, develop risk management guidelines to decrease exposure, and strengthen the ability to deliver opinions by cooperating with partners like JBCE and NIA. More details can be found in Annex B.

Austria: NanoTrust

One example of anticipatory governance in advanced materials is the NanoTrust programme and its role in the evolving nanotechnology risk governance ecosystem in Austria. Originating in 2007 as a short-term project, over time NanoTrust acted, and continues to act, as a keystone in the national ecosystem and as **a carrier and orchestrator of a programme of governance activities across multiple agencies and actors**. Since the early/mid-2000s, regulators and Environmental, Health and Safety (EHS) experts struggled with an abundance of uncertainties regarding the nature of nanomaterials and nanotechnologies and trying to keep up with the rapid developments in their research and development. Maintaining close ties with the Austrian nanotechnology governance system, NanoTrust has been able to have continuous interactions and consultation with EHS and risk expertise as well – allowing for an agile, and adaptable

anticipatory governance process that could react to the rapid evolution of nanotechnology and integrate anticipatory intelligence into its governance. As a key part of the Austrian nanotechnology governance system, NanoTrust has become almost an institution in and of itself, providing robust and independent information with a high degree of trust and legitimacy.

4 Reflection on strategic intelligence tools and approaches for advanced materials.

What is strategic intelligence?

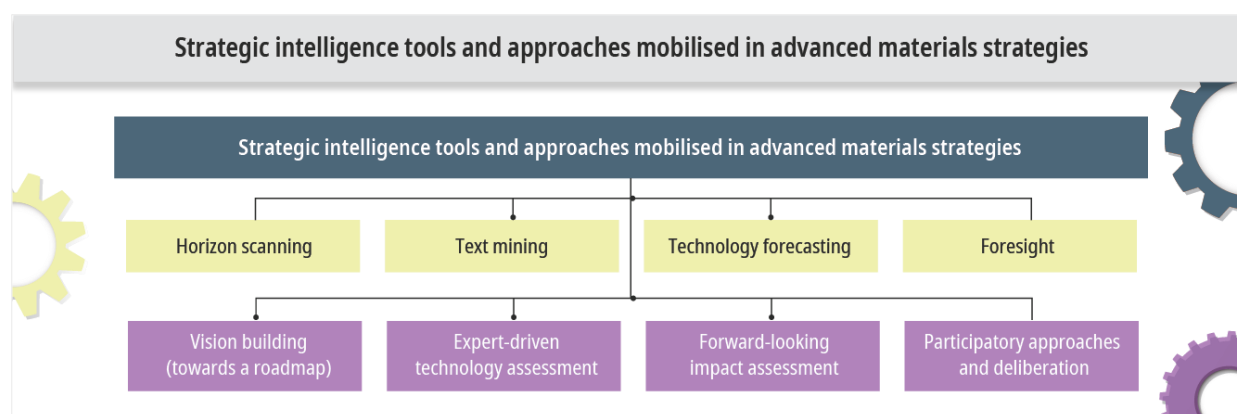
“Strategic intelligence” on the potential developments and implications of new and emerging technologies is a critical resource for building national strategies for emerging technology fields. Strategic intelligence refers to analysis and knowledge of the possible directions and economic stakes of its development, levels of societal support, possible ethical and societal aspects that may need to be considered, and potential impacts, benefits and risks. Methods include foresight, technology assessment, forecasting, and a number of other tools (a first introduction to these tools can be found in (Robinson and Doherty, 2025^[16])).

For national strategies on critical and emerging technologies, a number of strategic intelligence tools and approaches are mobilised. For advanced materials, the case studies included in this report incorporate different mixes of tools and approaches. The different approaches include:

- **Horizon scanning:** the search for weak signals of disruptive technologies (or other areas of interest) that may become impactful, or areas of concern, in the near future. With the speed of technological innovation increasing, the need for regular and robust horizon scanning is becoming increasingly important.
- **Text mining:** advances in both computation (speed and complexity), the availability of organised digital data sets that can be mined (scientific articles, patents, investment databases, to name a few) and the increasing number of policy-related data analysts, means that big data analytics can provide historical and near-real-time evidence of emerging technologies that can be mobilised for “near casting”.¹⁵ Text mining is a relatively mature example of big data analytics.
- **Technology Forecasting:** involves identifying trends in technology developments through mobilising multiple sources, and in some cases using modelling.
- **Foresight:** mobilising multiple techniques to imagine, in a systematic way, alternative futures, foresight draws on diverse and multidisciplinary perspectives to ensure robust and resilient assumptions about the future, helping to “future proof” governance of emerging technologies, underlying policy strategies and approaches. Foresight helps policymakers identify blind spots and potential future vulnerabilities with existing strategies as well as interconnections and opportunities for synergies between different actors or domains.
- **Vision building (towards a roadmap):** collective articulation of a vision to pursue is a key part of building national strategies. Identifying the shared expectations from key actors in the emerging technology domain (those that will develop, grow and deploy technologies) are translated into an agenda of action – embodied in a roadmap.

- **Expert-driven technology assessment:** involving deep-dive analysis and structured engagement with technology experts, expert-driven technology assessment brings together multiple knowledge sources to understand the potential developments of a technology and its ramifications for society, for example the societal, ethical, legal, and environmental issues as well as risks.
- **Forward-looking impact assessment:** builds off traditional technology assessment (where the technology being assessed has already materialised) and adds a further complexity of anticipating the future developments of a technology itself along with its broader ramifications. This is a key approach for early stage emerging (and potentially critical or highly impactful) technologies.
- **Participatory approaches and deliberation:** mobilises a set of tools, approaches and processes of assessing socio-technical issues by proactively including a variety of stakeholders in a process of deliberation and assessment. By bringing together different stakeholder groups, participatory approaches stimulate public and political opinion forming on societal and ethical aspects of technology and may also foster public trust through engagement and inclusion – particularly relevant for controversial technologies.

Figure 4.1. Strategic intelligence tools and approaches mobilised in advanced materials strategies



Source: OECD Research

Strategic intelligence mixes in advanced material case examples

To better understand the role of strategic intelligence in developing national strategies, it is worthwhile to unpack some of the case examples in the previous sections of the report. The 12 case studies described in previous sections, a number of which were further elaborated on by experts (see Annex B), provide a useful insight here.

Table 4.1. reveals that, even with the small sample of case studies described in this report, there are common areas of strategic intelligence: roadmapping, technology assessment and expert engagement. The other forms of strategic intelligence are mobilised in rather varied way. To help better understand this table, the strategic intelligence activities of three of the case studies will be briefly described: the National Materials Innovation Strategy (United Kingdom), PRIMA Quebec (Canada), the 3rd SIP and Materials Ecosystem Project (Japan). What is visible in these three case studies is that each strategic intelligence approach can provide a particular insight, but when used in combination with others, helps facilitate anticipatory processes and catalyse national strategy development.

Table 4.1. Use of strategic intelligence in report case studies

	Horizon Scanning	Text Mining	Technology Forecasting	Foresight	Vision building (towards a roadmap)	Expert-driven Tech. Assessment	Forward-looking impact assessment	Participatory approaches and RRI
PRIMA Quebec (Canada)	✓	✓	✓		✓	✓		
Materials Innovation Strategy (Japan)			✓		✓	✓		
3rd SIP and Materials Ecosystem Project (Japan)			✓	✓	✓	✓	✓	✓
Carbon Nanotubes (Japan)					✓	✓		✓
Strategic Agenda for Industry and Manufacturing (Portugal)			✓		✓	✓		
National Materials Innovation Strategy (United Kingdom)			✓		✓	✓		
AMMT Roadmap 2022 (United States)			✓		✓	✓		
Critical Minerals Technology Assessment (United States)						✓	✓	
Communication on Advanced Materials for Industrial Leadership 2024 (European Commission)			✓		✓	✓		✓
Safe and Sustainable by Design framework (European Commission)					✓	✓		
Early4AdMa (OECD)			✓			✓		
NanoTrust (Austria)			✓		✓	✓	✓	

Note: The Secretariat analysed the case studies presented throughout the report to extract their use of different strategic intelligence tools (marked as green when they were used, and red when they were not). Steering group delegates corroborated this analysis for their national examples and provided additional details where needed.

United Kingdom – National Materials Innovation Strategy

This case reveals a combination of strategic intelligence: forecasts for identifying trends, expert-driven technology assessment, participatory approaches and collective vision building to build the national strategy.

The overall aim of this activity was to build a shared vision where opportunities from current and future materials innovation offers to solve societal challenges and, at the same time, to realise new economic activity for the United Kingdom. Since the whole approach of the strategy-in-development was to be challenge-led, rather than technology-centric, the activity identifies the key national and global trends, **forecasts**, and drivers. Thus, a **vision building** exercise was set up (see section 3.1.2.) to develop a national coherency to provide focus to future opportunities and to support the business case for implementing the final Strategy's key recommendations which will include aspects such as funding support for the commercialisation of materials; the development of investment vehicles; greater incentives for companies to invest in materials innovation and support for skills programmes. The visioning exercise along with expert engagement (including **expert-driven technology assessment**) aimed to highlight any substantial gaps in national infrastructure in areas such as scale-up or verification. Under the stewardship of the Materials Innovation Leadership Group (MILG), the activity included a **participatory approach** consulting over 1200 materials innovators and worked closely with 350 actors across a wide range of strategic industries. Also, the Royce Institute which spearheaded this activity had open invitation for individuals to join or input to these expert working groups through submitting an Expression of Interest Form.

Canada – PRIMA Quebec

This case combines a variety of forms of strategic intelligence, text mining, horizon scanning, technology forecasting, expert-driven technology assessment, and a vision building exercise, elaborated below.

PRIMA Quebec undertook a **vision building** exercise to develop an “Advanced Materials 2035 Roadmap”. The activity mobilised key actors in the advanced materials ecosystem with a view to contributing to the decarbonization of the economy; further develop industry-research relationships to position Quebec in the marketplace, and better position the advanced materials sector as a pillar of Quebec's decarbonization objectives. The creation of the Roadmap 2035 allowed the collective envisioning of scalable solutions to the challenges surrounding the energy transition, the improvement of competitiveness and the security of supply chains. It was built in four key phases: 1) updates on global trends and ecosystem (via studies, surveys and interviews); 2) identification of strategic sectors and themes; 3) Vision 2035 workshop and preliminary definition of guidelines; 4) Co-drafting of objectives and actions (via working groups). It is comprised of 7 broad guidelines (e.g. promote SME growth, develop skilled workers) each broken down into more detailed objectives. This activity was supported by technology forecasting and horizon scanning **to identify international trends and weak signals of what may be coming** that may influence demand for advanced materials, which trends have been observed and retained in the 2018 report updates. Reported in the publication “*Advanced Materials: A Strategic Sector for Quebec*” report published in 2018, this technology forecasting and horizon scanning was mobilised to gain a better understanding of the industrial advanced materials ecosystem in the province of Quebec, factors that may be taken into consideration when envisioning its evolution, and to position advanced materials as a strategic sector essential to other industries.

In this process, the first step was to identify a series of international trends likely to strengthen and stimulate demand for advanced materials, followed by a series of needs and end markets for each trend. Next, an analysis of local demand was conducted via a survey of industry associations and research consortia actively working with an industrial clientele. Although by no means exhaustive, the analysis offered an initial overview of local perceptions with regard to key advanced materials, their impact and barriers to demand.

Expert-driven technology assessment activities were conducted. For example, in 2018, industry (including SMEs), government and academic stakeholders were engaged to identify the needs of industries and determine Quebec's competitive advantages, a questionnaire was sent to the 340 companies in the database, with questions about their advanced materials activities, company size and potential (innovation, exports, investment, growth), and whether they faced any barriers to their development. The approach adopted helped create an initial statistical portrait of the types of industrial expertise in Quebec, characterized by either the nature of the advanced materials involved, or the sectors of application served by the businesses in Quebec's industry. This activity also explored three major societal challenges for Quebec: 1) Complete the energy transition and decarbonize industry to achieve net zero by 2050 through collaboration with the innovation zones; 2) Increase productivity with a digital transition to help maintain and improve our standard of living; 3) Secure the country's strategic supply chains by reducing its dependence on foreign minerals, materials, components, and services needed to fuel the energy and digital transitions.

A key element of this activity included **text mining**: bibliometric analysis (2019, 2024) of scientific publications and patent requests for research and innovation in advanced materials in Quebec between 2003-2020. This provided key insights into the broad developments of advanced materials R&D over the previous five years. In particular, it identified the main and emerging research themes, as well as to what extent advanced material research is being guided by sustainable development values and calls for multidisciplinary collaboration. In parallel, a survey (N: 127) carried out as part of the Roadmap 2035 also provided an insight into the technological trends associated with the four application sectors that underpinned the choice of priority sectors.

Japan – 3rd SIP and Materials Ecosystem Project

This example offers a different mix of strategic intelligence approaches to PRIMA Quebec, though there are overlaps. The case includes forecasting, forward-looking technology assessment, expert-driven technology assessment, foresight and vision building and public engagement through consultation.

Central to this activity was a **vision building** exercise that identified a vision for the Japanese Materials Ecosystem. This then mobilised traditional **foresight** approaches, such as back-casting, to be able to define a variety of indicators, based on readiness levels. **Expert-driven technology assessment** was conducted to assess the factors that will shape the societal uptake of developed technologies, level of maturity of other various aspects such as governance, business, social acceptability, and human resources required as well as technologies themselves. This was done by using five indicators: Technology Readiness Level (TRL), Business Readiness Level (BRL), Governance Readiness Level (GRL), Social (Communal) Readiness Level (S(C)RL), and Human Resources Readiness Level (HRL). Each indicator is typically comprised of eight or nine sub-readiness levels which covers the phases from basic research to development and implementation. This was a combination of traditional technology assessment (looking at ramifications of existing technologies) and **forward-looking technology assessment**, which also anticipated on the development of the technologies themselves.

A key part of this activity was the exploration of macro trends through **forecasting**. A hundred potential markets trends in 2040 were first identified based on analysis of a number of macro trends (e.g., dynamics of population, climate changes) and expectations of key experts on future technology developments. The potential markets were screened by whether they were (a) materials-related, (b) had a threshold of market size >200 billion yen, and (c) whether the data-driven approach will be effective. A total of 25 markets remained after this screening, relating to applications such as of biotechnology, computing, social infrastructure, energy, and mobility. The conclusion was that research themes oriented to these markets would be a good way to drive forward economic benefits.

The developed R&D plan for project “*Creating a materials innovation ecosystem for industrialization*” was assessed through the public comment procedure as a means of **public engagement**.

5 Conclusions

Advanced materials, with their unique properties and performance advantages, promise to drive economic growth and help tackle societies' grand global challenges. For example, materials innovation can be used to build more sustainable and higher-performing products that reduce waste and carbon intensity to support the green transition. Countries, increasingly recognising them as a critical technology, are seeking to deliver the policy investment needed to accelerate their development and implementation.

This report analysed the **current policy practices in contemporary advanced materials strategy development and their strategic intelligence capabilities**. It first landscapes the national approaches pursued by different countries to foster advanced materials research and development, as well as the policy drivers shaping these efforts. The anticipatory processes that inform and enable these strategies (e.g. roadmapping, upstream governance) are unpacked and illustrated by a series of case studies from across the world. Lastly, the use of strategic intelligence tools (e.g. technology assessment, horizon scanning) across some of these examples is extracted to articulate the capacities needed to develop advanced material strategies.

With these insights, **the report informs efforts by decision-makers to deliver national strategies, policies or roadmaps to accelerate innovation, develop ecosystems and implement advanced materials solutions**. By analysing the anticipatory tools employed in existing activities and how strategic intelligence is generated and used, this report aims to support technology assessment practitioners in providing timely intelligence to their audiences to steer the direction of technology development.

What was not analysed in this report was the quality of the anticipatory processes and their outcomes, nor the quality of the strategic intelligence tools and outcomes that were part of these exercises. Moreover, whilst anecdotal evidence was provided on who undertook the strategic intelligence activities, it was not clear what are the local capacities and strategic intelligence producing ecosystems that the national strategies drew upon. These elements were beyond the scope of this study, but would be worthwhile for deeper investigation in a broader analysis of the role of strategic intelligence for critical and emerging technologies.

Annex A. List of national strategies, roadmaps or policies on advanced materials

Table A.1. National strategies, roadmaps or policies mentioning advanced materials across OECD members and accession countries

Country	Activity Name	Lead Organisation	Date
Argentina	National Science, Technology and Innovation Plan 2030	Ministry for Science, Technology and Innovation	Feb-23
Australia	List of Critical Technologies in the National Interest	Department of Industry, Science and Resources	May-23
	Advanced Materials and Manufacturing – Implications for Defence to 2040	Department of Defence	Sep-18
	Advanced Manufacturing Roadmap	CSIRO (Australia's National Science Agency)	Nov-16
Austria	Austrian Nanotechnology Action Plan	Federal Ministry of Agriculture, Regions and Tourism	Mar-09
Belgium	Wallonia's Smart Specialisation Strategy (S3) 2021-2027	Wallonia Regional Government	Mar-21
Canada	National Quantum Strategy	Ministry of Innovation, Science and Industry	Jan-23
	Next Generation Manufacturing Canada Advanced Manufacturing Supercluster Five-Year Strategy	Next Generation Manufacturing Canada	2022
Chile	Advanced Manufacturing Innovation Program	Advanced Manufacturing Innovation Program (IMA+)	Dec-18
Croatia	Research Infrastructure Development Roadmap of the Republic of Croatia 2023-2027	Ministry of Science and Education	Dec-23
	Smart specialization strategy of the Republic of Croatia	Ministry of Economy and Sustainable Development, Ministry of Science and Education	2016
Czechia	National Research and Innovation Strategy for Smart Specialisation of the Czech Republic 2021–2027	Ministry of Industry and Trade	Jan-23
Denmark	Strategy for Quantum Technology: Part 1	Ministry of Higher Education and Science	Jun-23
Germany	Position Paper Advanced Materials	German Environment Agency (Umweltbundesamt)	Mar-23
	Towards Safe and Sustainable Advanced (Nano)materials: A proposal for an early awareness and action system for advanced materials (Early4AdMa)	Ministry of Health, Welfare and Sport (RIVM), German Environment Agency (Umweltbundesamt), Federal Institute for Occupational Safety and Health (BAuA) and the Federal Institute for Risk Assessment (BfR)	May-22
	Principles paper on support for materials research	Federal Ministry of Education and Research (BMBF)	Feb-22
	Thematic Conferences Advanced Materials	German Environment Agency (Umweltbundesamt)	Jan-22
	Risk Governance of Advanced Materials	German Environment Agency (Umweltbundesamt), Federal Institute for Occupational Safety and Health (BAuA) and the Federal Institute for Risk Assessment (BfR)	Dec-21
	Advanced materials: Overview of the field and screening criteria for relevance assessment	German Environment Agency (Umweltbundesamt)	Jul-20
	Opportunities and Risks of Advanced Materials - Summary of the Discussions at the Expert Dialogue on Advanced Materials	Ökopol – Institute for Environmental Strategies	Aug-19
Iceland	Icelandic Roadmap for Research Infrastructures 2021	Ministry of Education, Science and Culture	Aug-21
	Iceland and the fourth industrial revolution	Government of Iceland, Prime Minister's Office	Oct-19
Ireland	Midlands Advanced Manufacturing Action Plan 2021-2024	Department of Enterprise, Trade and Employment	Jul-21

	Industry 4.0 Strategy 2020-2025	Department of Enterprise, Trade and Employment	Dec-19
Japan	R&D Strategy for Nanotechnology and Material Science and Technology	Ministry of Education, Culture, Sports, Science and Technology (MEXT)	Sep-24
	Vision for the new materials industry	Ministry of Economy, Trade and Industry (METI)	Apr-22
	Materials Innovation Strategy	Cabinet Office	Apr-21 (revised on Jun-25)
Korea	National Strategic Technology Nurture Plan	National Science Technology Advisory Board	May-23
	Strategy for Securing Future Materials to Support National Strategic Technologies	Ministry of Science and ICT	Mar-23
Lithuania	Smart Specialisation Strategy	Ministry of Economy and Innovation	Aug-22
	Lithuanian Industry Digitisation Roadmap 2019-2030	Ministry of Economy and Innovation	Mar-19
Luxembourg	National Research and Innovation Strategy	Ministry of Higher Education and Research	Feb-20
The Netherlands	Roadmap Chemistry of Advanced Materials	ChemistryNL	May-21
	Dutch Materials Agenda – Accelerating Materials Technologies	MaterialsNL	Mar-21
New Zealand	Advanced Manufacturing Industry Transformation Plan	Ministry for Economic Development	Mar-23
Poland	Polish Roadmap for Research Infrastructures	Ministry of Science and Higher Education	Oct-20
Portugal	National Smart Specialisation Strategy 2030	National Innovation Agency	Jun-22
Romania	Romanian Roadmap of Research Infrastructures	Ministry of Research and Innovation, Romanian Research Infrastructure Committee	Sep-17
Slovenia	Research Infrastructure Roadmap 2030 (NRR 2030)	Ministry of Education, Science and Sport	Apr-20
Spain	RDI Goals and Priorities - Materials for a sustainable future	MATERPLAT	Jun-21
	Spanish Science, Technology and Innovation Strategy 2021-2027	Ministry for Science and Innovation	2021
Switzerland	Roadmap for Research Infrastructures 2023	State Secretariat for Education, Research and Innovation (SERI)	2023
United Kingdom	National Materials Innovation Strategy	Henry Royce Institute	Jan-25
United States	National Strategy for Advanced Manufacturing	National Science and Technology Council	Oct-22
	Advanced Materials and Manufacturing Technologies (AMMT) - 2022 Roadmap	Department of Energy, Office of Nuclear Energy	Sep-22
	National Nanotechnology Initiative Strategic Plan	National Science and Technology Council	Oct-21
European Union	Advanced Materials for Industrial Leadership	European Commission	Feb-24
	Strategic Materials Agenda	Advanced Materials Initiative 2030	Apr-23
	Materials 2030 Roadmap	Advanced Materials Initiative 2030	Dec-22
	Materials 2030 Manifesto	Advanced Materials Initiative 2030	Feb-22

Note: The contents of this table served as the basis for Figure 2.1. Only OECD countries and candidate accession countries (as of November 2024: Argentina, Brazil, Bulgaria, Croatia, Indonesia, Peru, Romania, Thailand) were investigated. It does not claim to represent an exhaustive list, only what could be found via desk search.

Annex B. Case Studies

To deepen the analysis in this report and provide practical real-world evidence, the Secretariat identified and analysed a series of advanced materials national strategies, agendas, policies, communications and projects from across the world. Steering Group delegates supported these efforts by identifying additional examples from their national contexts and providing a more detailed write-up of these (see this Annex B), following guidance co-prepared between the Secretariat and the Steering Group (see Annex C).

1. Canada: Advanced Materials as a Strategic Sector for Quebec

Case study authors: Tyler Norsten (National Research Council Canada, Emerging Technologies) and Marie-Pierre Ippersiel (PRIMA Quebec)

Introduction

Quebec is Canada's largest province by area with a population of ~8.6 million, it comprises ~21% of Canada's population. Manufacturing has and continues to be an important economic sector for Quebec particularly in aerospace, mines & metallurgy, automotive, and technology industries. Advanced materials adoption and integration into the manufacturing sector is pivotal in this context as advanced materials have the ability to impart superior physical and function properties relative to the conventional materials they replace. However, the integration of advanced materials into existing manufacturing lines and processes is often challenging due to the inherent differences in the materials properties and processing. Furthermore, many traditional manufacturing sectors are unaware of advanced materials and the related manufacturing processes that are emerging. Recognizing the importance of advanced materials integration into the manufacturing sector, along with actors of the ecosystem, the province of Quebec created a non-profit organization in 2014 called PRIMA Quebec with a mission to support and facilitate the advanced materials ecosystem by promoting collaborative innovation in the province's materials and manufacturing sector to ensure that Quebec remains at the forefront of technological advancements, market demands and sustainable economic development.

Technology Scope: Advanced Material Specific? Or a broader strategy that includes Advanced Materials

The "Advanced Materials: A Strategic Sector for Quebec"¹ document published in 2018 by PRIMA Quebec outlines Quebec's strategic focus on developing and promoting advanced materials as a key economic driver. It emphasizes the province's strengths in research and innovation, its robust industrial base, and the collaborative efforts between academia, industry, and government. It was an essential tool for an initial categorizing and mapping of the ecosystem with an industry focus while improving the sector's market position and facilitating the implementation of public policies to better support advanced materials.

The document highlighted the potential for advanced materials to revolutionize various sectors, including aerospace, health, and energy, through enhanced performance and sustainability of advanced materials. It also underscored the importance of investing in talent, infrastructure, and international partnerships to maintain Quebec's competitive edge and foster economic growth.

Organizer/Orchestrator of the Activity: Lead & critical supporting organizers of the activity

PRIMA Quebec is a non-profit organization with a mission to support and facilitate the advanced materials ecosystem by promoting collaborative innovation for the economic development of Quebec. With a mandate to foster innovation, research, and collaboration among industry players, academic institutions, and government bodies it aims to support the development and commercialization of advanced materials, enhance the competitiveness of Quebec's industrial sectors, and stimulate economic growth by facilitating access to resources, expertise, and funding opportunities. PRIMA bridges the gap between research and market needs, ensuring the continuous advancement and application of cutting-edge materials technologies. PRIMA operates on funding from the provincial government (~65%) with the remainder coming from private funding generated through membership fees. Special initiatives such as this ecosystem/road-mapping initiative are typically supported through joint funding by provincial and federal innovation and economic development agencies with various stakeholders, including public research institutions, industry and academic partners taking part.

Type of Anticipatory Coordination: Is it an agenda setting initiative? Roadmap? Strategic plan for the construction of an innovation ecosystem or cluster?

Advanced Materials: A Strategic Sector for Quebec – 2018 edition, provided an initial synopsis of the industrial advanced materials ecosystem and its potential. It revealed a critical mass of ~340 companies that rely on research expertise to advance their business and become more competitive in their markets. It was intended to be a first edition to establish a baseline of the industrial advanced materials and supporting ecosystems in the province with follow-up reports updating key sector information while addressing identified challenges and gaps. It was essential tool for mapping and improving the sector's market position and facilitating the implementation of public policies to better support advanced materials in the province. It is worth noting that in a recent update of the initial survey, carried out in 2024 to support the Advanced Materials 2035 roadmap approach, the number of companies involved in the advanced materials ecosystem increased to 570, revealing better coverage of the sector. This demonstrates the industrial interest and/or adoption of advanced materials and/or processes.

Initiating Factors: What factors triggered the development of the anticipatory exercise?

Strategic intelligence in the form of technology potential, economic and market valuation reports identified Advanced Materials as an important growth market globally. According to Transparency Market Research, the advanced materials market was anticipated to be worth over US\$100 billion by the year 2024. This prospect echoed the findings of a McKinsey² report that listed advanced materials among 10 technologies likely to have the highest potential economic impact by 2025. According to current estimates the global market for Advanced Materials is ~US\$60, (2022), is projected to reach a revised size of US\$82 billion by 2028³.

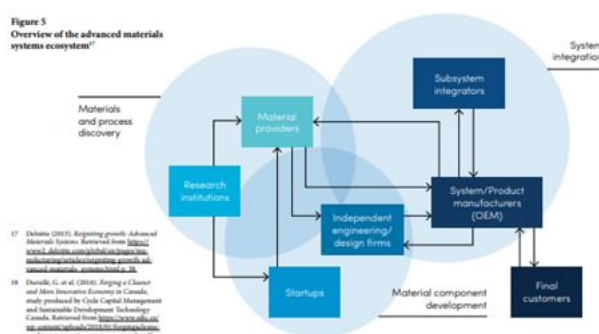
Although advanced materials were identified as strategic in advancing and enabling the innovation and manufacturing industry sector for the province, there was no recognized definition or single North American Industry Classification System for the sector. The cross-sector nature of advanced materials and their ability to be made into composites make characterizing the sector even more complex. With this information and the importance of materials and manufacturing to the GDP of Quebec, PRIMA set out to define the term «advanced materials» and then adapt the definition to Quebec realities.

The goal was to better define the industrial advanced materials ecosystem, identify stakeholders and their needs, and to create baseline metrics to identify research, innovation and policy gaps to better track future progress against targets.

Envisioned Outcome: What are the main objectives of the agenda?

The objective of the exercise in 2018 was to gain a better understanding of the industrial advanced materials ecosystem in the province of Quebec and to position advanced materials as a strategic sector essential to other industries. This portrait provided an initial overview of the booming sector by highlighting international trends that are influencing the demand for advanced materials. It revealed the challenges and opportunities present throughout the value chains, while demonstrating the undeniable potential and importance of these materials in the technological development of Quebec industries.

Figure B.1. Overview of the advanced materials systems ecosystems



Stakeholder Inclusion: Which stakeholders are included and in what way are they included? What insights are provided by stakeholder involvement?

PRIMA Québec, the sector-based association for industrial research, was the driving force behind the activity. As the preferred interface for research and development it convened industry, government and academic stakeholders to define and map the ecosystem. The primary stakeholders were SME companies with an interest in advanced materials ecosystem. They provided valuable information through targeted surveys identifying key material, product and process information to better classify (basic materials, semi-finished or finished products and/or processes and instrumentation) where they operate in the advanced materials/innovation spectrum.

Figure B.2. Advanced materials industrial ecosystem



Process: What are the different steps and activities? Methods & tools mobilized? What capacities are needed to successfully undertake the technology assessment approach?

A four-step approach was adopted to complete the mandate. It was critical to have PRIMA Quebec as the convener and organizer of the activity with support from the federal and provincial innovation agencies. An independent company, performing economic impact and cost-benefit analysis studies, in conjunction with a consulting firm, were selected to draw up a portrait of the Quebec advanced materials ecosystem with representatives of local innovation companies and institutions serving as advisors on the project's steering committee.

The first step was to identify a series of international trends likely to strengthen and stimulate demand for advanced materials, followed by a series of needs and end markets for each trend (Figure B.3.).

Figure B.3. General approach for the study 'Advanced Materials: A Strategic Sector for Quebec'



Next, an analysis of local demand was conducted via a survey of industry associations and research consortia actively working with an industrial clientele. Although by no means exhaustive, the analysis offered an initial overview of local perceptions with regard to key advanced materials, their impact and barriers to demand. To document the structure of the industrial ecosystem, a database of companies was created using information gathered from various sources.

To identify the needs of industries and determine Quebec's competitive advantages, a questionnaire was sent to the 340 companies in the database. A stratified sampling approach was used since it was an atypical group of businesses. Once the definition of the industry was stabilized—in other words, the qualification criteria of companies in the advanced materials industry—the starting point for the quantitative data research was prior knowledge of PRIMA Québec and its partner companies in the Quebec industry.

For the purposes of the study, this database was then enhanced: A survey was sent to these companies with questions about their advanced materials activities, company size and potential (innovation, exports, investment, growth), and whether they faced any barriers to their development. In addition, each company's Website was systematically consulted to identify the category or categories of advanced materials and the area or areas of application. Any reference to producing, integrating, or developing a process for an advanced material led to the inclusion of the company in these categories or fields.

The approach adopted helped create an initial statistical portrait of the types of industrial expertise in Quebec, characterized by either the nature of the advanced materials involved, or the sectors of application served by the businesses in Quebec's industry.

Frequency: Is this a one-off activity? Or repeated at regular intervals?

The initial activity spanned other activities and communications to further elucidate and document the advanced materials sector in Quebec: Advanced Materials – An Industry of Innovation and Prosperity (2021)⁴ – a further compilation of the 2018 report mapping advanced materials companies across the province and highlighting specific examples of advanced material applications to specific end-user markets such as transportation, construction, housing and energy. Advanced Materials – Quebec Innovation in Action – (2024) identified new companies that actively work in the advanced materials sector and updating the prospects and challenges and sharing the bibliometric analysis (publications⁶ and patents⁷) that was gathered as separate studies.

PRIMA Quebec has recently undertaken an Advanced Materials 2035 Roadmap exercise. The activity aims to mobilize the advanced materials ecosystem with a view to contributing to the decarbonization of the economy; further develop industry-research relationships to position Quebec in the marketplace, and better position the advanced materials sector as a pillar of Quebec's decarbonization objectives. The creation of the Roadmap 2035 aims to offer scalable and effective solutions to the challenges surrounding the energy transition, the improvement of competitiveness and the security of supply chains. Seven major orientations stem from this vision, which aims to position Québec and Canada on international markets while strengthening the strategic role of advanced materials for the future of our planet: 1) Increasing the circularity of advanced materials within supply chains, 2) Becoming a research leader, 3) Intensifying the emergence of start-ups, 4) Encouraging the growth of small and medium-sized enterprises, 5) Developing the next generation of skilled workers, 6) Ensuring more patient capital for companies, 7) Modernizing public procurement and regulatory frameworks.

Audience: Who is/are the target(s) for the intelligence that is produced by the technology assessments/roadmaps/strategies?

The primary audience for roadmaps, strategies and bibliometric reports is the industry and industry associations. PRIMAQuebec is able to use the data in the reports to drive policy and direct funding towards key advanced materials application areas for the province. The work provides evidence for both provincial and federal innovation agencies to create and direct funding programs to challenge and gap areas identified in the reports. The intelligence is informative for industry players in the ecosystem, providing connections to each other, funding, talent and supporting research organizations both academic and government.

Notes

¹ https://www.prima.ca/wp-content/uploads/2020/03/advanced_materials_prima_quebec_-june_2018.pdf

² McKinsey Global Institute (2013). Disruptive technologies: Advances that will transform life, business, and the global economy. Retrieved from https://www.mckinsey.com/~media/McKinsey/Business%20Functions/McKinsey%20Digital/Our%20Insights/Disruptive%20technologies/MGI_Disruptive_technologies_Full_report_May2013.ashx

³ Global Advanced Materials Industry – Industry Reports (precisionreports.co): GLOBAL ADVANCED MATERIALS INDUSTRY RESEARCH REPORT, GROWTH TRENDS AND COMPETITIVE ANALYSIS 2022-2028, SKU ID: QYR-21714951 | Publishing Date: 15-Sep-2022

⁴ <https://www.prima.ca/wp-content/uploads/2022/01/Document-promotionnel-ang-2.pdf>

⁵ https://www.prima.ca/wp-content/uploads/2024/06/PRIMA-24-109_Portrait_2024_EN_V2-1.pdf

⁶ [prima.ca/wp-content/uploads/2020/03/prima_quebec_rapport_2019_brevets_publications_vf.pdf](https://www.prima.ca/wp-content/uploads/2020/03/prima_quebec_rapport_2019_brevets_publications_vf.pdf)

⁷ https://www.prima.ca/wp-content/uploads/2024/05/SM_PRIMA_rapport_Executif_02-02-2024-1.pdf

⁸ https://www.prima.ca/wp-content/uploads/2024/09/2024-09-16_Feuille-de-route-2035_NonJustifie.pdf

2. Japan: Materials Innovation Strategy

Case study authors: Asako Okamura (NISTEP, MEXT) and Ayari Takamura (CRDS, JST)

Background

The Materials Innovation Strategy¹ was approved by the Council for Integrated Innovation Strategy at the Cabinet Office in April 2021 as one of the sectoral strategies in line with the 6th Science, Technology, and Innovation Basic Plan². Other sectoral strategies are formulated for the areas of artificial intelligence, biotechnology, quantum technology, fusion technology, health and medicine, space, ocean, and food, agriculture, forestry, and fisheries.

Under the Science, Technology, and Innovation Basic Law, the government is required to formulate the Science, Technology, and Innovation (STI) Basic Plan (for which the current 6th Plan covers the period from 2020 to 2025). The STI Basic Plan requires that the government to prepare an annual Integrated Innovation Strategy³. This Integrated Innovation Strategy is developed aligned with the sectoral strategies. The Integrated Innovation Strategy Secretariat leads the formulation and implementation of the sectoral strategies through cross-ministerial efforts.

Declining Japan's Competencies of the Material Industry and Science Base

With limited natural resources, Japan has had a strong industrial competitiveness in manufacturing, supported by a solid foundation of leading materials science. The strength in material science is the basis for its technology development and production capabilities, covering a wide range of products. These include general-purpose products, specialty function products, and functional materials. For years, Japan has held a significant share of the world market in these areas.

The country's strong scientific base in materials science has also led to groundbreaking discoveries. Two notable examples are blue light-emitting diodes and lithium-ion batteries. The developers of both these innovations were awarded Nobel Prizes, further highlighting Japan's significant contributions to the field of materials science.

In recent years, however, Japan has faced challenges in the materials industry. Its production share of some materials has declined due to several factors, including increased research and development in emerging markets, the commoditisation of products, and increased price competition. The industry is also grappling with supply chain security issues, which are largely due to the uneven geographical distribution of rare metals and rare earths that are essential for advanced electronic parts and functional chemical products. Given these challenges, securing sustainable competitiveness has become indispensable for Japan's manufacturing sector.

Academia in the materials science field also faces significant challenges. The number of research output has been declining, with a sharp drop in the international share of materials-related publications. In addition, the ageing of research talent has become a pressing issue. To address these issues, there is a growing need to improve the research environment to attract and retain younger researchers in the field.

Opportunities and challenges: Environments, Key Enabling Technologies and External Factors

Materials are increasingly expected to contribute to global sustainability goals, such as the Sustainable Development Goals (SDGs). This is in line with growing environmental and societal concerns.

New opportunities for innovative technologies have emerged, particularly in materials informatics and manufacturing process technologies. These key enabling technologies are driving progress in the field.

The COVID-19 pandemic has become an important catalyst for the development of the Materials Innovation Strategy. It has highlighted risks and vulnerabilities in international supply chains, requiring a robust strategy to ensure stable supplies and economic resilience.

A key challenge is to create an advanced innovation ecosystem that bridges the gap between basic research and societal implementation and ensures that new technologies developed in academia move towards commercialisation.

Formulating the Strategy

The Materials Strategy Expert Meeting was set up in October 2020 as a subsidiary body of the Council for Integrated Innovation Strategy. The Expert Meeting was composed of experts from academia (researchers, people with corporate experience, etc.) and industry (chemicals, steel, non-ferrous metals, etc.).

The preparatory report prepared by the Ministry of Economy, Trade and Industry (METI) and the Ministry of Education, Culture, Sports, Science and Technology (MEXT) served as the basis for the discussion at the Expert Meeting. The preparatory report focused on four key areas, including 1) establishment and improvement of a data-centric platform for materials R&D, 2) strategic promotion of key materials technologies and implementation areas, 3) creation of a materials innovation ecosystem, and 4) training and recruitment of human resources to support materials innovation capacity.

The Expert Meeting discussed i) technological trends (advances in materials informatics and manufacturing technologies), ii) environmental and societal needs (emphasis on the circular economy, resource recycling, and contributions to achieving the Sustainable Development Goals (SDGs)), iii) industrial and academic competitiveness (Japan's historical strengths and current challenges in materials development), and iv) international context (global competition for technological hegemony and the impact of international supply chain vulnerabilities highlighted by the COVID-19 pandemic). The expert group drafted the strategy document, which was finally approved in April 2021.

Technology Scope and the Structure of the Strategy

The strategy covers the whole field of materials science. In particular, the core technology area includes: i) advanced materials recycling technologies; ii) materials for innovative energy savings; iii) materials for advanced device performance; iv) materials with controlled quantum and electronic functions; v) next generation nanoscale materials; vi) next generation biomaterials and polymeric materials; vii) materials that function in extreme environments/uses; viii) multi-materialisation technology. For the enabling technology areas: ix) Design and control of materials and functions; x) Common enabling technologies for materials.

The strategy is structured as follows (See Figure B.4.). It has three pillars as 'basic guidelines': i) societal implementation; ii) data-driven research and development; iii) sustainable international competitiveness. Each guideline has several targets, and key performance indicators (KPI) are set for each target. Several initiatives (concrete policies) are linked to each target. Overall, the initiatives cover a wide range of policy areas, including: i) social implementation programmes; ii) research and development; iii) industry-academia-government cooperation; iv) human resource development; v) strategic international enterprise (including standardisation). Figure B.5. shows some examples of targets, KPI and concrete policies. The building materials commercialization innovation and cultivation ecosystem project in cross-ministerial strategic innovation promotion programs (SIP), which is introduced as the case study 3, is also linked to this strategy.

Figure B.4. Structure and components of the strategy

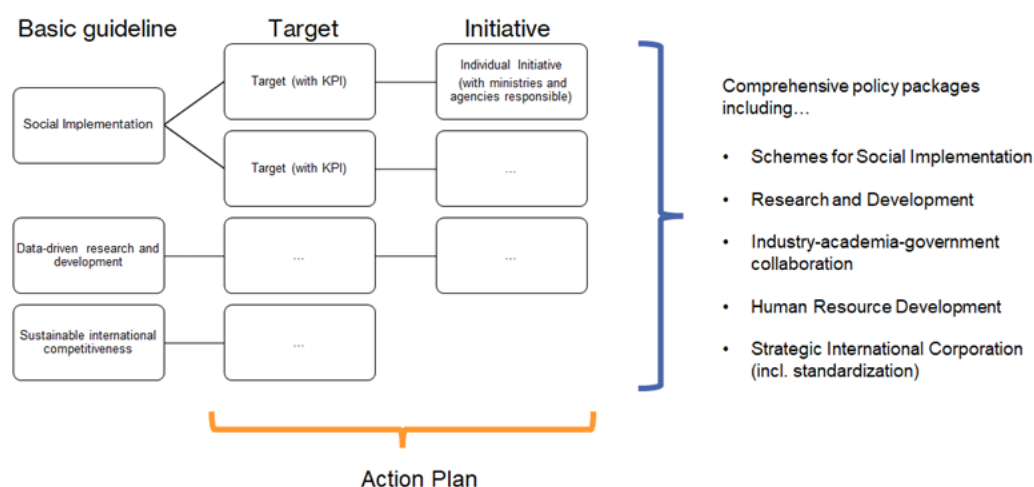
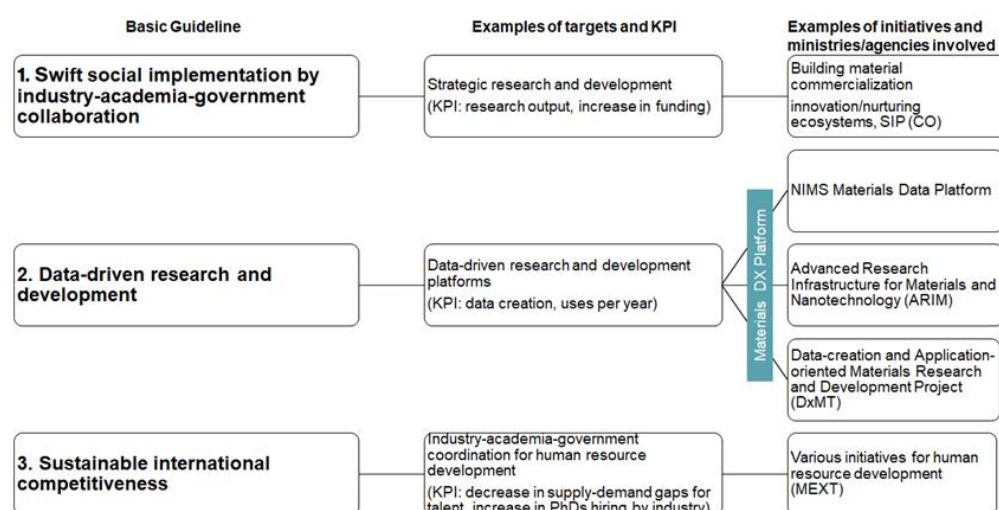


Figure B.5. Examples of initiatives linked to the strategy



Strategy Implementation

The Council of Integrated Innovation Strategy at the Cabinet Office leads the implementation of the sectoral strategies, including the Materials Innovation Strategy, in collaboration with relevant government agencies. The Secretariat examines and implements policies covering a wide range of aspects, including research and development, taxation, fiscal policy, institutions and regulation.

In terms of monitoring the strategy, the Materials Strategy Expert Meeting and its Task Force continuously monitor national and international trends, timely updates and adjustments, and promote stakeholder engagement. Discussions among relevant ministries and agencies have been underway to formulate the next strategy, which was published in June 2025⁴, in anticipation of the next STI Basic Plan, which is expected to start from 2026.

The Envisioned Outcome: Enhancing the National Potential for Materials Innovation

The envisaged outcome of the strategy should be to enhance the national potential to achieve "materials innovation" through coordination between academia, industry and government. This would make it possible to address societal issues and create new industries for the realisation of Society 5.0, develop markets that value reduced environmental impact, and create an innovation ecosystem that enables rapid social uptake. The strategy targets academia, industry, governments (including the Cabinet Office, MEXT, METI and Ministry of the Environment) and funding agencies (including JST, JSPS, NEDO).

Notes

1 Council for Integrated Innovation Strategy, Cabinet Office, Government of Japan (2021), "Materials innovation strategy,"

2 Government of Japan (2021), "The Sixth Science, Technology, and Innovation Basic Plan,"

3 Government of Japan (2024), "Integrated Innovation Strategy 2024,"

4 Council for Integrated Innovation Strategy, Cabinet Office, Government of Japan (2025), "Materials innovation strategy — Through the Construction of Knowledge Value Chains"

The newly revised Materials Innovation Strategy is available online in Japanese: https://www8.cao.go.jp/cstp/material/material_2025_honbun.pdf

3. Japan: Creating a Materials Innovation Ecosystem for Industrialization project in Cross-ministerial Strategic Innovation Promotion Programs (SIP)

Case study authors: Ayari Takamura (CRDS, JST) and Asako Okamura (NISTEP, MEXT)

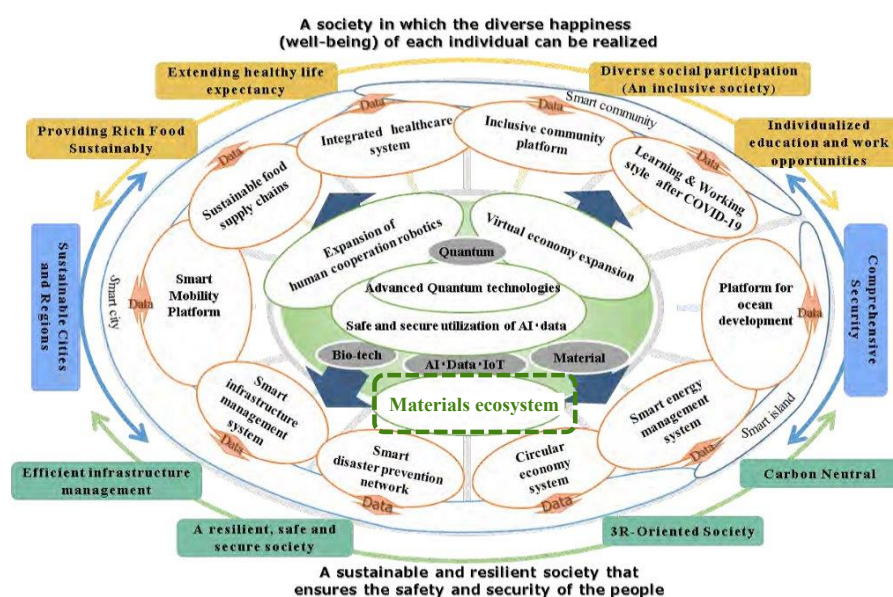
The 3rd period of SIP

Overview

Since 2014, under the leadership of Council for Science, Technology and Innovation (CSTI) at the Cabinet Office, Japan has been implementing a series of national programs called Cross-ministerial Strategic Innovation Promotion Programs (SIP). The SIP aims to address Japan's major social problems and contribute to the revitalization of the Japanese economy, beyond the existing boundaries of government ministries and traditional disciplines. The SIP focus on end-to-end R&D from basic research to commercialization (social implementation).

After the 1st period (from 2014 to 2018) and the 2nd period (from 2018 to 2022), the 3rd period of SIP has been launched since 2023 under the target vision of "Society 5.0"¹ presented in the 6th STI basic plan. A total of 14 projects was selected by back-casting from the vision of Society 5.0. One of the projects entitled "Creating a materials innovation ecosystem for industrialization" (the materials ecosystem project) was adopted as one of the common fundamental technologies, along with quantum technology, biotech, and AI, also surrounded by other vision oriented projects (Figure. B.6.).

Figure B.6. Overview of the 3rd period of SIP and Materials ecosystem project positioned as common fundamental technologies



Redesign of the 3rd SIP

The framework of the 3rd SIP period was redesigned by referring to some opinions and issues collected through the mid-term evaluation of the 2nd SIP, reports from the program directors (PD), a proposal by economic organization, and interviews to the inside of Cabinet Office, related ministries, and the program management institutes (PM). Then some points for anticipatory coordination were indicated: Need to narrow down to specific themes by back-casting from the future vision shown in the STI basic plan; the definition of “social implementation (of technology)” need to be clear. Accordingly, the target areas in the 3rd SIP were identified via back-casting from the vision of “Society 5.0”. In addition, “Readiness levels” toward social implementation were set not only for an aspect of technology (TRL) but also for governance, business, social acceptability, and human resources.

Readiness levels (indicators) for social implementation

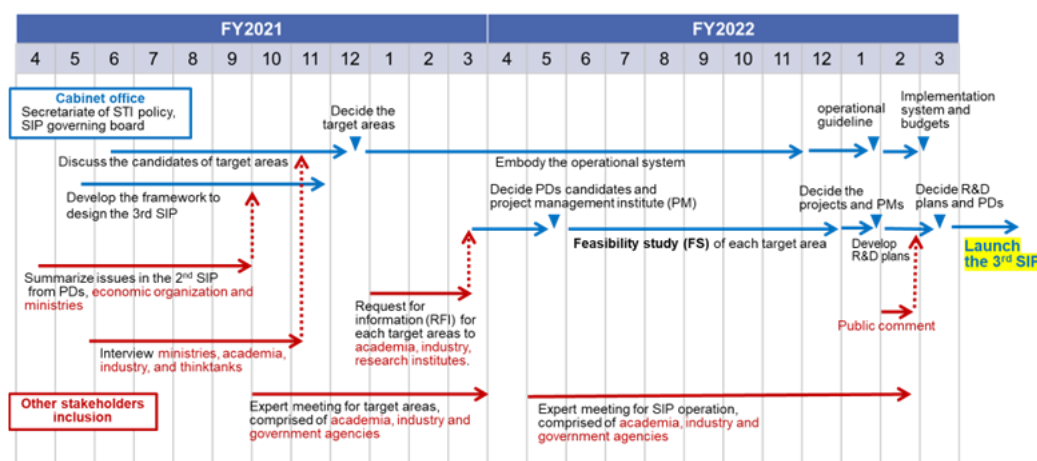
To realize social implementation of developed technologies, maturation of other various aspects such as governance, business, social acceptability, and human resources are required as well as technologies themselves. Whereas conventional program had barely promoted the other aspects except technologies, the 3rd SIP defined the readiness levels in terms of five essential perspectives: Technology Readiness Level (TRL), Business Readiness Level (BRL), Governance Readiness Level (GRL), Social (Communal) Readiness Level (S(C)RL), and Human Resources Readiness Level (HRL). Each indicator is typically comprised of eight or nine readiness levels which covers the phases from basic research to development and implementation.

Process of designing the 3rd SIP

The process to design the 3rd SIP was mainly organized by Secretariate of Science, Technology and Innovation Policy in Cabinet Office, and the SIP governing board. Also, opinions of the CSTI expert members were incorporated. During FY2021, the framework to design the SIP (e.g., back-casting from “Society 5.0”, operational ways to promote end-to-end R&D) was discussed. Also, opinions on potential

target areas were gathered from various sectors and screened through Request for Information (RFI). Subsequently, in FY2022, feasibility studies (FS) for each target area (resulted in 14 adopted out of 15 candidates) were conducted and the respective R&D plans were developed as “projects”. Details of the process and stakeholder inclusion are summarized in Figure B.7.

Figure B.7. The process and stakeholder inclusion during designing the 3rd period of SIP



A project “Creating a materials innovation ecosystem for industrialization”

Background

Materials is key industry of Japan which occupies almost one third of the GDP generated by manufacturing industries. Recently, however, the global competitiveness of materials industry of Japan has been decreasing. This is considered due to lower R&D efficiency. Japanese companies have hardly increased R&D efficiency for recent ten years, compared to other global companies.

A promising approach to increase the R&D efficiency is a data-driven development of materials. During the 1st and 2nd SIP, the materials-related projects (i.e., “Structural Materials for Innovation” project in the 1st SIP, “Materials Integration for Revolutionary Design System of Structural Materials” project in the 2nd SIP) had been dedicated to developing “materials integration systems”². The “materials integration systems” aimed to reduce costs and shorten development periods by combining material engineering methods and data science. The data-driven approach of materials development has been expected for various application fields. The approach should be promoted and then lead to strengthening the competitiveness of materials industry.

Missions of the project

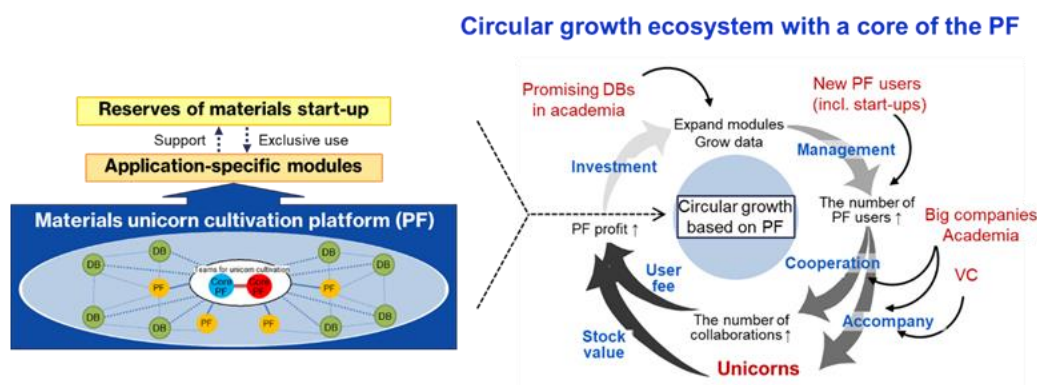
The materials ecosystem project aims to contribute to realizing the envisioned Society 5.0 via innovation of materials industry in Japan. The mission is described as follows,

- Develop ecosystems based on “Materials unicorn cultivation platform”
- Generate numerous materials companies with high R&D efficiency (materials unicorn)
- Grip the core network of materials process-data for social implementation

“Materials unicorn cultivation platform (PF)” acquires various types of data physically or creates them online, connects various individual data infrastructures existing in Japan. Application-specific modules will be constructed on the PF, which enable materials start-ups to exclusively utilize the data-infrastructures.

An ecosystem should be built with a core of the PF. Integration and expansion of existing data-infrastructures in academia promotes to generate and grow the start-ups. Simultaneously, the user fee and increases of the stock value grow the data-infrastructures. Thus, circular growth of the ecosystem will be established (Figure B.8.).

Figure B.8. Images of materials unicorn cultivation platform and circular growth ecosystem

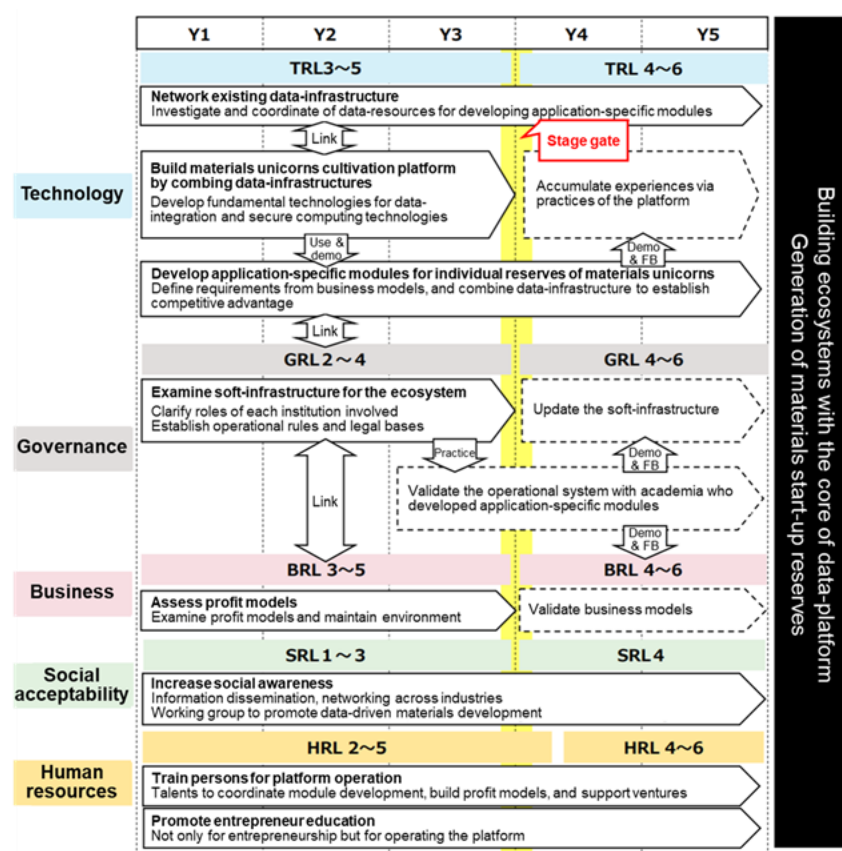


Roadmap and scenario for ecosystem building

According to the grand design of the 3rd SIP, a roadmap was developed for the perspectives of five indicators (i.e., TRL, BRL, GRL, S(C)RL, and HRL). Goal images were firstly defined, and eight or nine readiness levels were set via back-casting approach. The target levels during the 3rd period of SIP were determined, and then the roadmap was formulated (Figure B.9.). Moreover, along with the roadmap, three phases for establishing the ecosystem were expected as follows,

- Phase 1: During the 3rd SIP = Build the bases of ecosystems (2023-2027)
- Phase 2: Develop the ecosystems via practices (2028-2032)
- Phase 3: Establish the circular growth (2032-2040, Goal)

Figure B.9. A roadmap during the 3rd period SIP



Macro-trend analysis for targeting potential markets

Another emphasis in developing the program was to consider macro trends. One hundred of potential markets to be trends in 2040 were firstly identified based on macro trends (e.g., dynamics of population, climate changes) and expectation of technical improvements. The potential markets were screened by whether materials-related, a threshold of market size >200 billion yen, and whether the data-driven approach will be effective. A total of 25 markets remained were related application of bio, computing, social infrastructure, energy, or mobility. Hence, research themes oriented to these markets are supposed to be adopted in the project.

Process for launching the project

The FS was conducted to assess the following three points,

1. Comprehensive investigation of existing materials data-infrastructures in Japan
2. Trials of scenario making for materials unicorns
3. Macro-trend analysis for targeting potential markets

The R&D plan was subsequently established based on the results of the FS. In addition, three sub-projects for advancing the R&D and strategies for dissemination and global collaboration were presented. The developed R&D plan was assessed through the public comment procedure, and finally decided.

Launched in April 2023, the project invites research proposals at sub-project level and provides grants for projects proposed by academia and industry.

Notes

¹ (1) a sustainable and resilient society that ensures the safety and security of the people, and (2) a society in which the diverse happiness (well-being) of each individual can be realized.

² Such as MInt (Materials Integration by network technology) for structural metallic materials and CoSMIC (Comprehensive System for Materials Integration of CFRP) for carbon fiber reinforced plastics (CFRP).

4. Japan: Toward regulation of carbon nanotube: activity of an industry association, NBCI

Case study authors: Ayari Takamura (CRDS, JST) and Asako Okamura (NISTEP, MEXT)

Carbon nanotube: its broad applications and structural variety

Carbon nanotube (CNT) is a category of nano-scale tubular-shape carbon substances that contribute widely to a variety of applications, including batteries, memories, transmission cables, composite materials and medical/health cares. CNT is expected to be one of the key advanced materials for realizing sustainable ecosystems and individual well-being.

Such a wide range of applications for CNT depends on a great variety of structural features, including length, diameter, density, single- or multi-walled, surface property, rolled angle, morphology (e.g., tangled, cross-linked, junction), and carbon purity. These structural features are linked to the characteristic physiochemical properties of each CNT product.

The regulation of CNT, like other nano materials, is difficult to characterize and risk assess due to their diverse structures and properties. This situation is quite different from chemical compounds, which are defined with unique structural formulas.

Process toward CNT regulation in Europe

The evaluation of CNT, especially multi-walled CNT (MWCNT), had been started by Federal Institute for Occupational Safety and Health (BAuA) in Germany under Community Rolling Action Plan by European Chemicals Agency (ECHA), which was reported in 2019¹. Followingly, BAuA submitted a CLH report on MWCNT to ECHA for consideration toward CLP (classification, labeling and packaging) regulation in March 2021. The report proposed that MWCNT with diameter range ≥ 30 nm to <3 μ m and a length ≥ 5 μ m and an aspect ratio $\geq 3:1$, should be classified as Carcinogenic Category "Carc.1B" (presumed carcinogenic) in the CLP classification. Through public consultation and discussion in the Committee for Risk Assessment (RAC-60) in ECHA, the proposal was adopted as a RAC opinion in September 2022. Currently, the proposal is under discussion in the meeting of Competent Authorities for REACH and CLP (CARACAL), involving RAC-60 in ECHA, and DG GROW and DG ENV in European commission². Moreover, European Commission has issued a WTO/BTB Notification on MWCNT based on the BAuA proposal in January 2024.

Activity of NBCI along the process in Europe

NBCI is an industry association for nanotechnology business and the infrastructure in Japan, which is composed of more than 170 companies. CNT is one of the most important substances in Japanese industry including the application fields. Thus, the Social Acceptance and Standardization Committee in NBCI has contributed to disseminate information about evaluation of CNT to other institutes and communities in Japan and abroad.

Along the process of considering CLP regulation of WNCNT based on the BAuA proposal, NBCI submitted their comments to ECHA three times in August 2021³ (during public consultation), July 2022⁴ and August 2022⁵. The comments mainly argued the following:

- Need more scientific evidence about the possibility of carcinogenic. All CNT products cannot be judged equally only by the diameter and length.
- Regard the IARC ((International Agency for Research on Cancer)) classification: most of CNT are classified in Group3 of “not classifiable”, except MWNT7 in Group 2B of “Possibly carcinogenic”. Then, harmonize with international rules.
- Also need to develop risk management methods for broad applications of CNT.

After the RAC opinion was adopted, NBCI, cooperated with Japan Business Council in Europe (JBCE), has continuously proposed an alternative classification criterion in the 48th CARACAL meeting in March 2023, and presented some scientific evidence such as a carcinogenicity study of MWCNT⁶ in the 50th CARACAL meeting in November 2023. JBCE also issued a position paper to remind the opinion of the Japanese industry in April 2024.

Moreover, NBCI and Nanotechnology Industries Association (NIA) signed a Memorandum of Understanding (MoU) for cooperation in responding to the regulation at the end of 2023. NIA presented their opinions, based on the NBCI’s comments, to the Nano expert meeting of ECHA in April 2024.

NBCI will continue the activity to realize both innovation and safety based on scientific evidence of MWCNT. NBCI is specifically planning to,

- Execute research to elucidate the mechanism of possible harmfulness of CNT, especially investigate potential factors such as remaining catalyst, crystallinity, and surface properties.
- Develop a guideline for risk management to decrease exposure to CNT.
- Strengthen the ability to deliver opinions by cooperating with European associations such as JBCE and NIA.

Notes

1 SUBSTANCE EVALUATION CONCLUSION and EVALUATION REPORT for Multi-Wall Carbon Nanotubes (MWCNT), synthetic graphite in tubular shape and tangled (EC No 936-414-1/701-160-0)

2 On September 30, 2024, European Commission released the revised CLP regulation which ensures to label MWCNT with diameter range ≥ 30 nm to <3 μm and a length ≥ 5 μm and an aspect ratio $\geq 3:1$, as “Carc.1B”.

3 [http://www.nbci.jp/file/220203_MWC\(N\)T_CLH_en_p7.pdf](http://www.nbci.jp/file/220203_MWC(N)T_CLH_en_p7.pdf)

4 http://www.nbci.jp/file/NBCI_Comment_on_CLH_report_20220708.pdf

5 https://www.nbci.jp/file/NBCI_Comment_on_CLH_report_20220817e.pdf

6 Sato, K., et al., Difference in carcinogenicities of two different vapor grown carbon fibres with different physicochemical characteristics induced by intratracheal instillation in rats. *Particle and Fibre Toxicology*, 20(1), 37 (2023).

5. Portugal: Strategic Agenda for Industry and Manufacturing

Case study authors: Luis Humberto Viseu Melo, (IST, Technical University of Lisbon)

The **Strategic Agenda for Industry and Manufacturing** is an Agenda assessing technologies and technology trends for Industry and Manufacturing with a 2030 horizon. Five main technology areas were identified, and tackled from an application point of view. Technologies were also approached considering their maturity (research or innovation stage).

Technology scope

The agenda has the objective of identifying technologies crucial or relevant for Industry and Manufacturing in the medium term. This assessment should provide stakeholders with information helping to shape development strategies, both for the research and the industry communities.

Advanced Materials was one of the five key areas identified. Their potential interest for industrial development was thoroughly assessed. It should be noted that this assessment work was performed by experts from multiple stakeholder origins (academia, industry, government) on a collaborative, voluntary basis.

Central organizer / orchestrator of the agenda

The Agenda was initiated by the FCT (Fundação para a Ciência e a Tecnologia), the Research Council from the Ministry handling Science within the Portuguese Government. The FCT provided coordination and technical support to the experts working on the agenda.

Two Coordinators were appointed from research and innovation communities.

Type of anticipatory coordination

The Agenda could be placed between a broad roadmapping and a foresighting activity. Academic and Industry actors could use this information to build their own roadmaps and strategies.

Initiating factors

A need was identified to provide roadmapping and foresighting information to the industry community in general, and in particular the Advanced Materials community. It was important to provide these stakeholders with a source of information helping them to plan their strategies. A very broad view with a 2030 horizon is very important for this, and was not available to most.

Another objective was to foster dialogue between Industry, Academia and Public actors. This dialogue has traditionally been difficult to establish for many reasons (different cultures, mismatch between the research produced and the industry needs), and this exercise was very successful from this point of view. In the case of Advanced Materials this gap was very obvious.

Envisioned outcome

The main intended outcome of this Agenda goes from shaping the research strategies to incorporating innovative technologies into Industry and Manufacturing.

Another was to foster collaboration between actors working at different TRL levels and diminishing gaps in knowledge chains.

Stakeholder Inclusion

The Agenda was created by a group of 40 experts with two coordinators. These volunteered from a larger pool of invited experts. These came from Research (Academia, research Institutions) and the private sector (Industry), and there were also experts from FCT.

Process

In a first step a very large group of experts (around 120) were invited for very broad meeting. From these 40 volunteered to actually do the work, and two coordinators (one for research and one for innovation) were assigned. Of these 40, 8 provided expertise in Advanced Materials.

The experts held regular meetings, and the documents provided by each expert were then edited and integrated by the coordinators.

The experts were asked to provide knowledge/opinions on:

- Which technologies, or sets of technologies, should be relevant for Industry and Manufacturing with a 2030 horizon
- Foreseeable impact (economic, social, functional) in Industry with the same horizon
- Needs for the development of these technologies

Frequency

One-off activity, but to be revisited as needed.

Target Audience

Whole ecosystem (Academia and Research Community in general, Industry, Public Bodies, Regulators, Public)

Barriers

Other organisations (unions, consumer associations, etc.) did not show an interest to participate. Yet, the Agenda can be useful for these groups to understand the development of Industry and Manufacturing technologies in general, and Advanced Materials in particular.

6. United Kingdom: National Materials Innovation Strategy (NMIS)

Case study author: Thanuja Galhena (Department for Business and Trade)

Description

The Henry Royce Institute in the UK, facilitate the development of a National Materials Innovation Strategy (NMIS). This strategy, which will set out a 10-year vision, is pivotal to delivering a coherent approach for UK Materials sector, across Government, industry and the wider materials technology community.

Technology scope

The vision is to establish a national strategy for materials innovation which delivers coherency through the government, industry, academia and the materials community, giving the UK a world-leading position in the rapidly expanding multi-billion-pound materials markets, substantially enhancing the value of UK-based industries.

It will reflect the world-renowned strength of UK research in material science, the highly skilled and innovative materials industries in the UK, and the central role materials advances play in a truly innovative economy. It will further recognise that the UK's strengths in materials research and innovation span many

different sectors. This is significant given the overwhelming need for effective cross-sector collaboration to drive the wider growth, efficiency and productivity of the UK's economy.

The strategy will seek to grow and further develop our materials industry (both developers and users), ensure that this capability remains strongly rooted in the UK and deliver long-term benefits to the regional and national economy.

Initiating factors and envisioned outcome

This strategy is designed to ensure that Materials are clearly positioned as not only a strategic sector in its own right, but as a critical driver of innovation across a range of important technologies.

Past efforts to bring such cohesion to materials innovation activities have been often centred on sectors, and have not always been able to consider the synergies and opportunities in the links between industries and national policy drivers. Royce mentioned in its interim report “We have now reached the point where the UK will not be able to achieve its ambitions for a healthy and resilient nation operating in a prosperous net zero economy, in the timescales demanded, without the support of a specific, clear, and standalone national strategy for materials innovation and technology translation. Critically, there is a need for a strategy that is sustained over the five to ten years needed to deliver this change.”

This work will identify high impact areas where materials innovation will create significant value whilst addressing national priorities – from getting to net zero and rebalancing the UK economy, to supporting national resilience and enabling healthy lives. Also, it will develop a national coherency to provide focus to future opportunities, it will also support the business case for implementing the final Strategy's key recommendations which will include aspects such as funding support for the commercialisation of materials; the development of investment vehicles; greater incentives for companies to invest in materials innovation and support for skills programmes. The report will also highlight any substantial gaps in national infrastructure in areas such as scale-up or verification.

Central organizer / orchestrator of the agenda

The Henry Royce Institute is the UK's national institute for advanced materials research and innovation. Royce is a partnership of nine leading institutions spread across the UK. Royce has appointed an expert team comprising ScotCHEM, Perspective Economics and Urban Foresight to engage stakeholders across the materials science community in the development of the National Materials Innovation Strategy. ScotCHEM is the project lead, and, with the team, they have expanded on the Strategy Framework, working across the materials community to develop the strategy.

The report was developed under the leadership of the Materials Innovation Leadership Group (MILG) formed following the Royce launch of the Strategic Framework for Action.

Stakeholder Inclusion

Royce initiated the strategy development emphasising a demand-led approach, accelerating technology translation and adding value to the UK economy. Following extensive engagement across all sectors, their interim report summarises more than 1,200 industry responses, which have been grouped into priority materials innovation areas for deeper exploration by the UK materials community. Also, Royce had open invitation for individuals to join or input to these expert working groups through submitting an Expression of Interest Form.

Through workshops and consultations, they have identified 6 “Core Themes” (Energy materials, soft materials, Biocompatible materials, Structural materials, Materials for surface enhancement, Materials for electronics, telecoms, sensing & computing technologies) and 6 “Cross-over Themes” (Sustainability,

Materials 4.0, Skills and education, Supply chain resilience, Scale-up capabilities, Regulations and standards) that the strategy should focus on.

Thirty Opportunity Workstreams have been identified across these Core Theme headings. They are currently convening experts and undertaking a deeper exploration of materials innovation in each priority area through Expert Working Groups.

Also, this work includes a comprehensive econometric output - the analysis has identified 2,574 companies in the UK, active in materials innovation, contributing just under £45 billion to the UK economy.

Target Audience

Whole ecosystem (HMG, Industry sectors, investors, exporters, funding bodies, technology developers, skill developers, regulatory and standard bodies).

7. United States: U.S. Government Accountability Office Assessment of Critical Minerals Recovery Technologies

Critical minerals such as lithium, cobalt, and rare earth elements are essential for technologies used in the military and the broader economy, including airplanes, rechargeable batteries, computers, cell phones, electrical systems, and advanced electronics, but these minerals are vulnerable to supply-chain disruptions. Domestic, nontraditional sources of minerals have been identified as a potential mechanism to reduce reliance on foreign suppliers and the need to open new mines, among other benefits.

Critical minerals are often produced from hardrock mines, though it is possible to produce them from nontraditional sources. The assessment identified several available and developing technologies for recovering critical minerals from nontraditional sources, including mining wastes, wastewater from mines, waste from coal-fired power plants, and saline water from underground formations (geothermal brines). For example, with the use of sorbent-based direct lithium extraction, lithium ions in a geothermal brine adhere to the surface of solid material called a sorbent while other types of ions may stay in the brine. This technology could offer advantages over the conventional process for recovering lithium from brines which uses evaporation, such as a smaller footprint and faster lithium recovery.

Several challenges to recovering critical minerals from nontraditional sources were also identified, including

- **Liability.** Recovery operations on previously mined sites could result in operators being responsible for historical liabilities. There is little appetite in industry to take on this financial risk, according to experts.
- **Economics.** Due to factors such as high fixed costs and unstable prices potential recovery project operators may be uncertain that their investments will be financially viable.
- **Public engagement and tribal consultation.** Stakeholders and experts identified engagement with local communities, and when appropriate, government-to-government consultation with Tribal Nations as important steps to a successful critical mineral recovery project.

Three policy options were identified by the TA practitioners that could help address the challenges or enhance the benefits of recovering critical minerals from nontraditional sources. Additionally, the report described ongoing legislation and deferral and private sector projects that could enhance U.S. efforts related to critical mineral recovery from non-traditional sources. If policymakers find these efforts to be sufficient, or if they find other sources of critical minerals more suitable to their goals they may choose to sustain current efforts by maintaining the status quo.

Table B.1. Policy options set out in GAO’s report “Critical Minerals: Status, Challenges, and Policy Options for Recovery from Nontraditional Sources”

Policy Option	Opportunities	Considerations
Pilot Good Samaritan legislation Implementation approaches: Legislators could provide some liability protections for third parties recovering critical minerals from waste at previously mined sites. Legislators could provide protections for third parties but require a portion of profits generated be used for restoration activities.	• Could encourage investment in domestic recovery operations. Could expand the types of organizations interested in cleaning up previously mined sites.	• Disturbing previously mined sites may result in new environmental effects. If financial assurances are not adequately set, federal or state taxpayers may become responsible for cleaning up additional environmental liabilities. Requiring that a percentage of profits be used for restoration activities could affect industry interest in previously mined sites.
Subsidies Implementation approaches: The federal government could subsidize the development of specific nontraditional sources to meet defense or energy needs. Subsidies could be made available as tax credits for those pursuing nontraditional sources of critical minerals.	• Properly tailored subsidies could catalyze technology development, demonstration, commercialization, and domestic production of critical minerals. Subsidies could help offset some of the fixed costs associated with developing recovery and processing infrastructure.	• Taxpayer-funded subsidies do not guarantee that supported recovery operations would become profitable. Once in place, subsidies can be difficult to end. May result in reallocation of resources from other priorities.
Community benefit agreements Implementation approaches: To improve engagement with communities near nontraditional sources, permitting agencies could encourage operators to pursue agreements that outline how communities may benefit from projects that also incur costs in their communities. Companies could adopt policies that encourage or facilitate these agreements.	• Negotiating specific community benefits from new recovery operations could create deeper acceptance of facilities that may have environmental effects. New recovery operations could offer additional employment opportunities in economically depressed communities.	• Negotiating which stakeholders benefit, which do not, and who controls the agreement can be challenging. It is difficult to predict who will be willing to engage in such agreements. Such agreements may be time consuming to create. Some provisions in these agreements may be difficult to enforce.
Status quoImplementation approach: Sustain current efforts.	• Federal policymakers could observe and evaluate existing efforts, such as agency funding of characterization of nontraditional sources, which could limit risks and resources expended. Continued private sector efforts, such as recovery of lithium from geothermal brines, could ultimately result in profitable recovery of minerals. The private sector may also pursue other options to overcome critical mineral supply chain issues. For example, buyers may pursue substitutes, reducing the need for new sources.	• Current efforts may not address all the challenges described in this report. Current efforts alone may delay or inhibit the development of nontraditional sources for critical minerals, which could result in forgone benefits such as increased independence from foreign suppliers.

Source: GAO | GEO-24-106395 (US GAO, 2024_[17]).

Annex C. Guiding questions for case study development

Table C.1. Guiding questions for analysis of national strategies on advanced materials

Theme	Key questions	Examples
Technology scope	Advanced Material specific? A broader strategy which includes advanced materials (i.e. advanced manufacturing strategy)?	Range of advanced materials linked to specific sectors (Royce National Materials Innovation Strategy) All novel materials (Japanese Materials Innovation Strategy - MEXT).
Central organizer / orchestrator of the agenda	Who is the lead organizer of the agenda? Which Ministry	Ministry, Public Research Organization, Parliament, Intermediary Organization?
Initiating factors	Why was the National Strategy initiated? What are the key policy drivers behind the strategy?	Industry led initiative, like the UK National Materials Innovation Strategy or public private partnerships like the Dutch Roadmap - Chemistry of Advanced Materials (see Table 1). Ministry led Japanese Materials Innovation Strategy (MEXT).
Envisioned outcome	What are the main objectives of the agenda? To build an innovation ecosystem? To enhance international cooperation?	Enhance Japan's material innovation capacity to strengthen competitiveness and address social issues (MEXT example, see Table 1).
Stakeholder Inclusion	Which stakeholders were included in developing the strategy? Who helped construct the strategy (active role) and who informed the strategy (passive role)?	Firms and industry associations, research communities, civil society organizations, citizens, user groups, trade unions, financial actors, regulatory agencies etc.
Target Audience	Who is the target for agenda? A specific Ministry? Whole of Government?	Whole ecosystem, national agency, ministry. industry consortia, etc.

Table C.2. Guiding questions for analysis of anticipatory coordination for building new and resilience advanced materials value chains

Theme	Key questions	Examples
Technology scope	Advanced Material specific? A broader strategy which includes advanced materials (i.e. advanced manufacturing strategy)?	Range of advanced materials linked to specific sectors (Royce National Materials Innovation Strategy) Part of Manufacturing Agenda (Portuguese example)
Central organizer / orchestrator of the activity	Who is the lead organizer of the agenda?	Ministry, Public Research Organization, Parliament, Intermediary Organization? Industry Association? Consortium?
Type of anticipatory coordination	Is it an agenda setting initiative? A roadmap? Strategic planning for the construction of an innovation ecosystem or cluster?	Roadmapping activity. Visioning exercise. Foresighting future sectors. Industry Scenarios.
Initiating factors	What factors triggered the development of the anticipatory exercise? How do they set/define 'problems' or 'situations' to solve/improve?	To develop a national strategy and roadmap to grow the advanced material ecosystem (UK example). To create coherence within the innovation

		ecosystem (Portuguese example).
Envisioned outcome	What are the main objectives of the agenda?	Roadmap, Strategy, Community Building, etc.
Stakeholder Inclusion	Which stakeholders are included? What insights are provided through stakeholder involvement? In what way are stakeholders included? What type of scientific disciplines are collaborating?	Firms and industry associations, research communities, civil society organizations, citizens, user groups (e.g. patient associations), trade unions, financial actors, regulatory agencies etc.
Process	What are the different steps and activities? Which methods and tools are mobilized? Which capacities are needed to successfully undertake the technology assessment approach?	Stakeholder engagement, horizon scanning, data mining, scenario development, constructive technology assessment, co-design, impact assessment.
Frequency	Is it a one-off activity? Or a repeated at regular intervals?	Punctual, quarterly, yearly, biennial?
Audience	Who is the target for the intelligence that is produced from the technology assessment?	Parliament, specific Ministries, Civil Society (for example to build trust), Firms (for example to stimulate responsible innovation), etc.
Barriers	What are the barriers or obstacles to be recognized or foreseen?	Difficult to identify and encourage consumer organisations or retailers to participate.

Table C.3. Guiding questions for anticipatory governance of advanced materials

Theme	Key questions	Examples
Technology scope	Advanced Material specific? A broader strategy which includes advanced materials (i.e. advanced manufacturing strategy)?	Nanomaterial focus (NanoTrust example) Range of advanced materials linked to specific sectors (Royce National Materials Innovation Strategy) All novel materials (Early4AdMa)
Central organizer / orchestrator of the agenda	Who is the lead organizer of the agenda?	Ministry, Public Research Organization, Parliament, Intermediary Organization? Industry Association? Consortium?
Type of anticipatory governance activity	Which form of anticipatory governance? Responsible Research and Innovation (RRI)? anticipatory Life-Cycle Analysis (LCA)? emerging risk assessment? Safe and Sustainable by Design (SSbD)? Other approach?	Responsible Research and Innovation (RRI), anticipatory Life-Cycle Analysis (LCA), emerging risk assessment, Safe and Sustainable by Design (SSbD).
Initiating factors	What factors triggered the anticipatory governance activity? How do they set/define 'problems' or 'situations' to solve/improve?	Concern of nanotoxicity (NanoTrust) Recognition of need to drive material innovation with an anticipatory screening approach (Early4AdMa)
Envisioned outcome	What are the main objectives of the activity	Insights into safety, risks, societal impacts. For example, the German example (Table 1), aims to promote advanced materials for circularity and energy transition. Integrate SSbD into innovation process.
Stakeholder Inclusion	Which stakeholders are included? What insights are provided through stakeholder involvement? In what way are stakeholders included?	Firms and industry associations, research communities, civil society organizations, citizens, user groups (e.g. patient associations), trade unions, financial actors, regulatory agencies etc.
Process	What are the different steps and activities? Which methods and tools are mobilized? Which capacities are needed to successfully undertake the technology assessment approach? What type of scientific disciplines are collaborating?	Stakeholder engagement, horizon scanning, data mining, scenario development, constructive technology assessment, co-design, impact assessment.
Frequency	Is it a one-off activity? Or a repeated at regular intervals?	Punctual, quarterly, yearly, biennial?
Audience	Who is the target for the intelligence that is produced from the technology assessment?	Parliament, specific Ministries, Civil Society (for example to build trust), Firms (for example to stimulate responsible innovation), etc.
Barriers	What are the barriers or obstacles to be recognized or foreseen?	Fewer incentives for academia to participate, etc.

Notes

¹ OECD (2023). Advanced Materials: Working Description. Series on the Safety of Manufactured Nanomaterials. No. 104. [https://one.oecd.org/document/ENV/CBC/MONO\(2022\)29/en/pdf](https://one.oecd.org/document/ENV/CBC/MONO(2022)29/en/pdf)

² https://taxation-customs.ec.europa.eu/carbon-border-adjustment-mechanism_en

³ For integrated circuits, or any other semiconductor device, a large number of process technologies and technology components are needed. As industry grew in this area, there was increased specialization on various component development and process technology improvement. Therefore, to progress and improve performance of devices, and to ensure that returns on investment of developing a single component of a device wasn't held up by delays in other components, the International Technology Roadmap for Semiconductors was initiated to coordinate across the different specialized firms, large and small, setting the pace of change. The technology roadmap therefore, helps identify when a certain component or capacity will be needed in a semiconductor device, and each developer can target the collectively agreed to date for their part of the overall device. https://www.semiconductors.org/wp-content/uploads/2018/09/1_Executive-Summary.pdf

⁴ More information is accessible online: https://www.royce.ac.uk/content/uploads/2025/01/National_Materials_Innovation_Strategy_FINAL_WEB_SITE.pdf

⁵ Whilst not directly on advanced materials, advanced and alternative process technologies for critical materials was part of this activity, and this case provides an interesting technology assessment component.

⁶ 82 Fed. Reg. 60835 (Dec. 26, 2017). This order was updated in 2020; see also Exec. Order No. 13953, Addressing the Threat to the Domestic Supply Chain From Reliance on Critical Minerals From Foreign Adversaries and Supporting the Domestic Mining and Processing Industries, 85 Fed. Reg. 62539 (Oct. 5, 2020)

⁷ Department of Commerce, A Federal Strategy to Ensure Secure and Reliable Supplies of Critical Minerals (Washington, DC: June 2019).

⁸ The U.S. GAO defines an open recommendation as one in which actions to satisfy the intent of the recommendation have not been taken or are being planned. See GAO, Critical Minerals: Building on Efforts to Advance Recovery and Substitution Could Help Address Supply Risks, GAO-22-104824 (Washington DC, June 16, 2022).

⁹ GAO, Critical Minerals: Status, Challenges, and Policy Options for Recovery from Nontraditional Sources, GAO-24-106395 (Washington DC, July 31, 2024).

¹⁰ <https://irgc.org/risk-governance/emerging-risk/a-protocol-for-dealing-with-emerging-risks/>

¹¹ Mazri, C. and M. Florin (2015), IRGC guidelines for emerging risk governance: guidance for the governance of unfamiliar risks., <https://infoscience.epfl.ch/record/228053>.

¹² <https://www.oecd.org/chemicalsafety/safer-and-sustainable-innovation-approach/early-awareness-and-action-system-for-advanced-materials-pre-regulatory-anticipatory-risk-governance-tool.pdf>

¹³ As a sidenote, the Working Party for Manufactured Nanomaterials (WPMN) was invited to present their approach and was very interested in learning more about anticipatory approaches stemming from the work in Module 1.2. We plan on continuing the exchange between the Steering Group and the WPMN.

¹⁴ Original Early4AdMa approach by the Netherlands' RIVM (National Institute for Public Health and the Environment) and Germany's BfR (German Federal Institute for Risk Assessment), BAuA (Federal Institute for Occupational Safety and Health) and UBA (German Environment Agency): https://www.rivm.nl/sites/default/files/2022-07/RIVM-brochure-Towards-safe-%28nano%29materials_V2_TG.pdf

¹⁵ "Near-casting" is a form of forecasting that extrapolates from near-real-time data developments in the near-future (0-3 years in the future).

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